Village Shoppes Suite 13

95 Washington Street, #222
Canton, MA

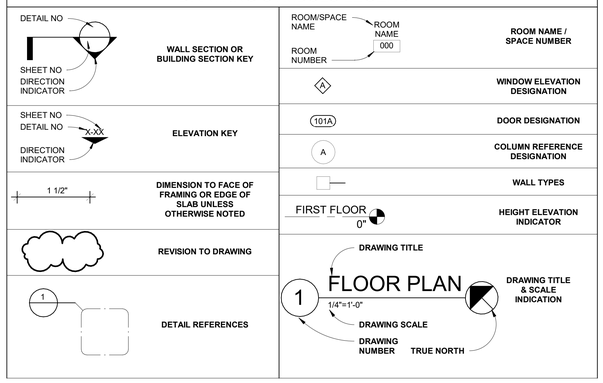
Building Permit
07/02/2026



INDEX OF DRAWINGS

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GENERAL				
G-001	COVER SHEET	1	Permit	07/02/26
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ARCHITECTURE				
A-101	FLOOR PLANS	1	Permit	07/02/26
A-121	ROOF PLAN	1	Permit	07/02/26
A-301	SECTIONS	1	Permit	07/02/26
A-201	ELEVATIONS	1	Permit	07/02/26
ELECTRICAL				
E-001	ELECTRICAL LEGEND, NOTES & FIXTURE SCHEDULE	1	Permit	07/02/26
E-100	ELECTRICAL FLOOR AND ROOF PLANS	1	Permit	07/02/26
E-200	ELECTRICAL SPECIFICATIONS	1	Permit	07/02/26
MECHANICAL				
M-001	MECHANICAL LEGEND, NOTES & SCHEDULES	1	Permit	07/02/26
M-101	MECHANICAL FLOOR PLANS	1	Permit	07/02/26
PLUMBING				
P-001	PLUMBING LEGEND, NOTES & DETAIL	1	Permit	07/02/26
P-101	PLUMBING FLOOR PLANS	1	Permit	07/02/26
P-201	PLUMBING SPECIFICATIONS	1	Permit	07/02/26

DRAWING SYMBOLS LEGEND



ABBREVIATIONS

ABBR	DEFINITION	ABBR	DEFINITION
ABV	ABOVE	HM	HOLLOW METAL
AC	AIR CONDITIONER	INT	INTERIOR
ACC	ACCESSIBLE	MAX	MAXIMUM
ADJ	ADJACENT	MBM	METAL BUILDING MANUFACTURER
AFF	ABOVE FINISH FLOOR	MECH	MECHANICAL
ALUM	ALUMINUM	MFR	MANUFACTURER
ARCH	ARCHITECTURAL	MIN	MINIMUM
BFF	BELOW FINISH FLOOR	MISC	MISCELLANEOUS
BLW	BELOW	MO	MASONRY OPENING
BO	BOTTOM OF	MTL	METAL
BRG	BEARING	NIC	NOT IN CONTRACT
CJ	CONTROL JOINT	NTS	NOT TO SCALE
CL	CENTERLINE	OC	ON CENTER
CLR	CLEAR	OPP	OPPOSITE
CMU	CONCRETE MASONRY UNIT	PLAM	PLASTIC LAMINATE
CONT	CONTINUOUS	PT	PRESSURE TREATED
DIA	DIAMETER	REF	REFERENCE
DWG	DRAWING	RO	ROUGH OPENING
EJ	EXPANSION JOINT	RTU	ROOF TOP UNIT
EL/ELEV	ELEVATION	SF	SQUARE FEET
ELEC	ELECTRICAL	SFRO	STOREFRONT ROUGH OPENING
ETR	EXISTING TO REMAIN	SIM	SIMILAR
EXT	EXTERIOR	STL	STEEL
EQ	EQUAL	STRUC	STRUCTURAL
FE	FIRE EXTINGUISHER	TO	TOP OF
FF	FINISH FLOOR	UNO	UNLESS NOTED OTHERWISE
FFE	FINISH FLOOR ELEVATION	THK	THICK
FRP	FIBERGLASS REINFORCED PLASTIC	TYP	TYPICAL
GA	GAUGE	VIF	VERIFY IN FIELD
GC	GENERAL CONTRACTOR		
GPS	GLASS PANEL SYSTEM		

PROJECT DIRECTORY

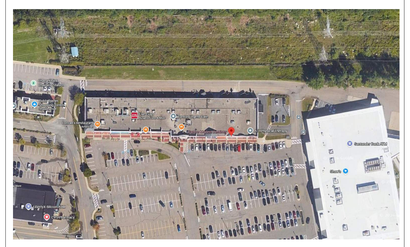
OWNER: **KIMCO REALTY**
2 Executive Park Drive
Suite #100
Newton, MA, 02462
(617) 686-4808
Joseph Polizzotti

ARCHITECT: **SPARC**
43 Valley Road
Needham, MA, 02492
(508) 284-6638
Aksel Solberg

STRUCTURAL ENGINEER: **VEITAS ENGINEERS**
315 University Ave, 2nd FL.
Westwood, MA, 2090
(860) 460-5800
Jonathan Bayreuther

MECHANICAL, ELECTRICAL, & PLUMBING ENGINEER: **BER**
66 Main Street
North Easton, MA, 02356
(508) 230-0260
Richard Fitts

BUILDING OFFICIAL: **Canton, MA**
801 Washington Street
Canton, MA, 02021
(781) 821-5003
Edward Walsh



N
VICINITY MAP

sparc

Sparc Design, PLLC
43 Valley Road
Needham, MA, 02492

t: (617) 905-4043

Client:
KIMCO REALTY

2 Executive Park Drive
Suite #100
Newton, MA, 02462

t: (516) 869-2951

Consultant:

Village Shoppes Suite 13
95 Washington Street, #222
Canton, MA 02021
JOB NUMBER: 2026021

ISSUE BLOCK

1	Permit	07/02/26

Seal:

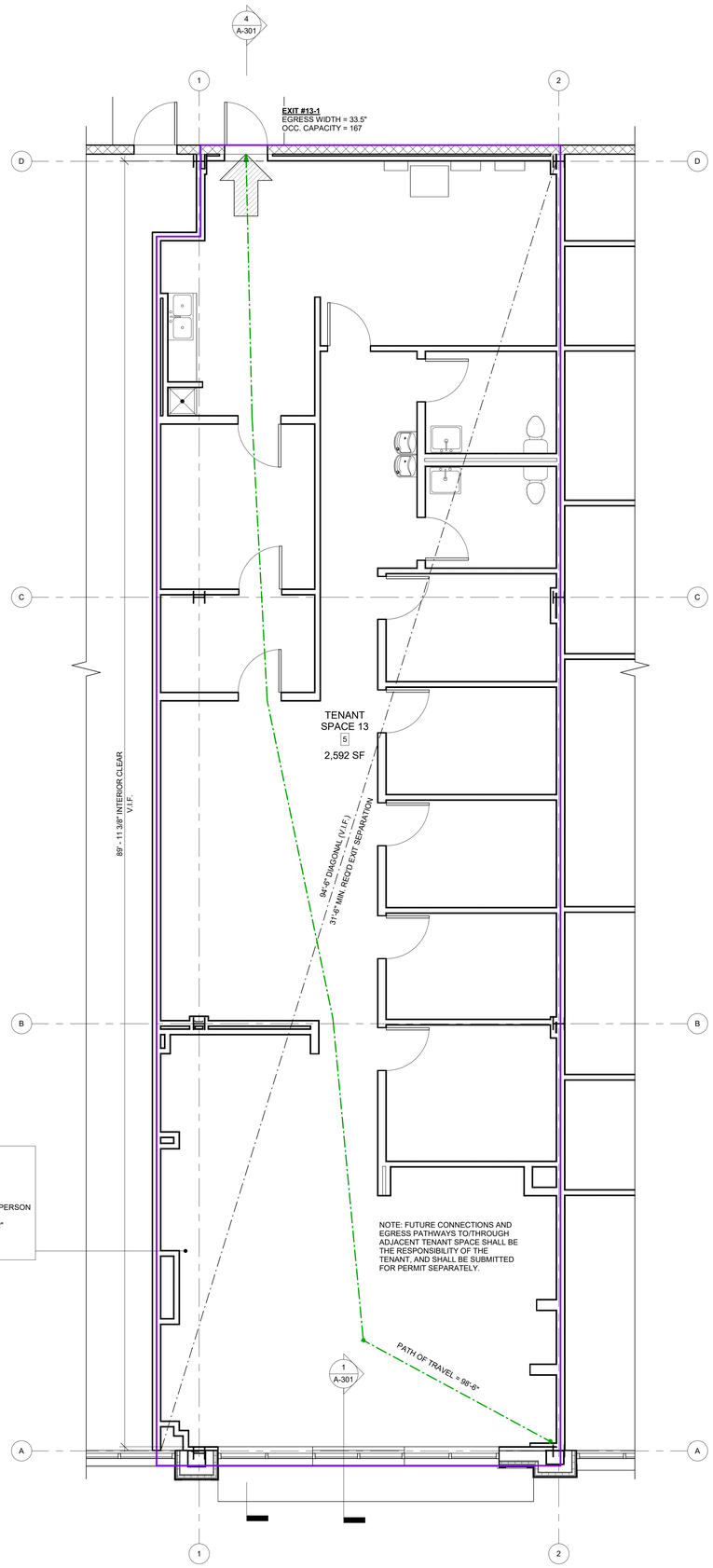


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DOCUMENT DATE: 07/02/2026

COVER SHEET

SHEET:
G-001

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G-001-COVER SHEET.pdf Parent



TENANT SUITE #13 - B - BUSINESS
2,592 SF

EGRESS REQUIREMENTS:

OCCUPANT USE: B - BUSINESS
OCCUPANT LOAD FACTOR: 150 SF/PERSON
OCCUPANT LOAD: 17
MIN. REQUIRED EGRESS WIDTH: 32"
EGRESS WIDTH PROVIDED: 33.5"
NO. OF EXITS REQ'D: 1
NO. OF EXITS PROVIDED: 1

1 G-101 OVERALL LIFE SAFETY FLOOR PLAN
1/4" = 1'-0"

VILLAGE SHOPPES SUITE 13 CORE & SHELL RENOVATION - CANTON, MA

Massachusetts Code Analysis

Applicable Building Code	780 CMR: Massachusetts State Building Code, 10th Edition, Chapter 34
Applicable Structural Code	780 CMR: Massachusetts State Building Code, 10th Edition, Chapter 34
Applicable Accessibility Code	521 CMR: Architectural Access Board Rules and Regulations
Applicable Plumbing Code	248 CMR 10.00 Massachusetts State Plumbing Code
Applicable Mechanical Code	780 CMR: Massachusetts State Building Code, Chapter 28
Applicable Fire Code	527 CMR 1.00: Massachusetts Board of Fire Prevention Regulations, NFPA 1
Applicable Electric Code	527 CMR 12.00: Massachusetts Electrical Code, NFPA 70
Project Summary	Demolition of existing interior tenant fit-out (finishes, fixtures, fittings, non-load bearing / non-rated walls, doors, hardware) to restore vacant shell condition suitable for future fit-out & expansion by Tenant in suite 14. Includes removal of existing Suite 13 storefront entrance and new openings in demising wall between suites 13 and 14 to facilitate future connections by Tenant. Future interior fit-out design and construction documents shall be submitted separately by the Tenant.

[Insert Version] International Existing Building Code (IEBC): Classification of Work & Level of Alteration

Section 503, Level 2 Alteration

Level 2 Alterations, as described in Section 603, shall comply with the requirements of Chapter 8 of the IEBC.

General Building Characteristics

Floor Plan Area (First Floor Footprint)	**2,592 SF** (Existing to Remain)
Number of Floors	**1** (Existing to Remain)
Total Square Footage	**2,592 SF** (Existing to Remain)

Chapter 3: Use and Occupancy Classification

Section 309: B Business (Existing to Remain)

Chapter 4: Special Detailed Requirements Based on Use and Occupancy

N/A

Chapter 5: General Building Heights and Areas

Allowable Building Height in Feet Above Grade Plane (Table 504.3)	N/A - Existing to Remain
Allowable Number of Stories Above Grade Plane (Table 504.4)	N/A - Existing to Remain
Allowable Area Factor in Square Feet (Table 506.2)	N/A - Existing to Remain

Chapter 6: Types of Construction

Construction Classification (Section 602.[Insert Section])	Type 2B (Existing to Remain)

Chapter 7: Fire and Smoke Protection Features

N/A

Chapter 8: Interior Finishes

Interior Wall and Ceiling Finish Requirements by Occupancy (Table 803.13)	Interior exit stairways and ramps and exit passageways	B
	Corridors and enclosure for exit access stairways and ramps	C
	Rooms and enclosed spaces	C

Chapter 9: Fire Protection Systems

1. **Section 903.2.7 Group M:** An automatic sprinkler system exists throughout the building, and shall remain. Existing sprinkler heads located in existing drop ceilings shall be adjusted to have upright heads and suitable for future modifications (as/if applicable) by Tenant. Any future adjustments to the existing fire protection system shall be the responsibility of the Tenant(s) and shall be submitted for permit separately.

1. **Section 907.2.7 Group M:** A fire alarm system exists in the building and shall remain. Any future adjustments to the existing fire protection alarm shall be the responsibility of the Tenant(s) and shall be submitted for permit separately.

Chapter 10: Means of Egress

Occupant Load (Section 1004)	**See Life Safety Plan(s) for occupant load calculation.**	
Means of Egress Sizing (Section 1005)		
Minimum Required Egress Width per Occupant	Doors/Other (Section 1005.3.2)	0.2" per occupant (min 32" clear per Section 1010.1.1) **See egress calculations on Life Safety Plan for additional information.**
	Stairways (Section 1005.3.1)	0.3" per occupant (min 44" per Section 1011.2) **See egress calculations on Life Safety Plan for additional information if applicable.**
Exit Access Travel Distance (Table 1017.2)	Occupancy B, With Sprinkler System	300 feet, **See Life Safety Plan for conformance.**
Exit and Exit Access and Doorway Configuration (Section 1007.1.1)	Where two exits, exit access doorway, exit access stairways or ramps, or any combination thereof, are required from any portion of the exit access, they shall be placed a distance apart equal to not less than one-half the length of the maximum overall diagonal dimension of the building or area to be served measured in a straight line between them. Where a building is equipped throughout with an automatic sprinkler system in accordance with section 903.3.1.1 or 903.3.1.2, the separation shall not be less than one-third of the length of the maximum overall diagonal dimension of the area served. **See Life Safety Plan for conformance.**	
Dead Ends (Section 1020.4)	Where more than one exit or exit access doorway is required, the exit access shall be arranged such that there are no dead ends in corridors more than **20 feet** in length. (Exception: In occupancies in Groups B, E, F, I-1, M, R-1, R-2, R-4, S and U, where the building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, the length of the dead-end corridor shall not exceed **50 feet**. **See Life Safety Plan for conformance.**	

Chapter 11: Accessibility

Design and construction of buildings shall comply to the accessibility standards outlined in Chapter 11, or the Accessibility Codes governing the state in which the project exists.

A building, facility, or element that is altered shall comply with Section [Insert correct section] of the IEBC.

LIFE SAFETY LEGEND

- PATH OF TRAVEL
- EXIT SEPARATION
- EXIT

FIRE EXTINGUISHERS

FE-1 = RECESSED FIRE EXTINGUISHER CABINET EQUAL TO LARSENS ARCHITECTURAL SERIES VERTICAL DUO IN A #4 STAINLESS STEEL FINISH

FE-2 = FIRE EXTINGUISHER ON WALL MOUNTED HOOK

FINAL LOCATIONS SHALL BE REVIEWED WITH THE FIRE MARSHALL, OWNER, AND ARCHITECT PRIOR TO INSTALLATION.

FIRE EXTINGUISHER TYPE/SIZE SHALL BE DETERMINED BY FIRE MARSHALL AND AHJ

MAX. TRAVEL DISTANCE OF 75' REQUIRED.

sparc

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t: (516) 869-2951

Consultant:

Village Shoppes Suite 13
95 Washington Street, #222
Canton, MA 02021
JOB NUMBER: 2026021

ISSUE BLOCK		
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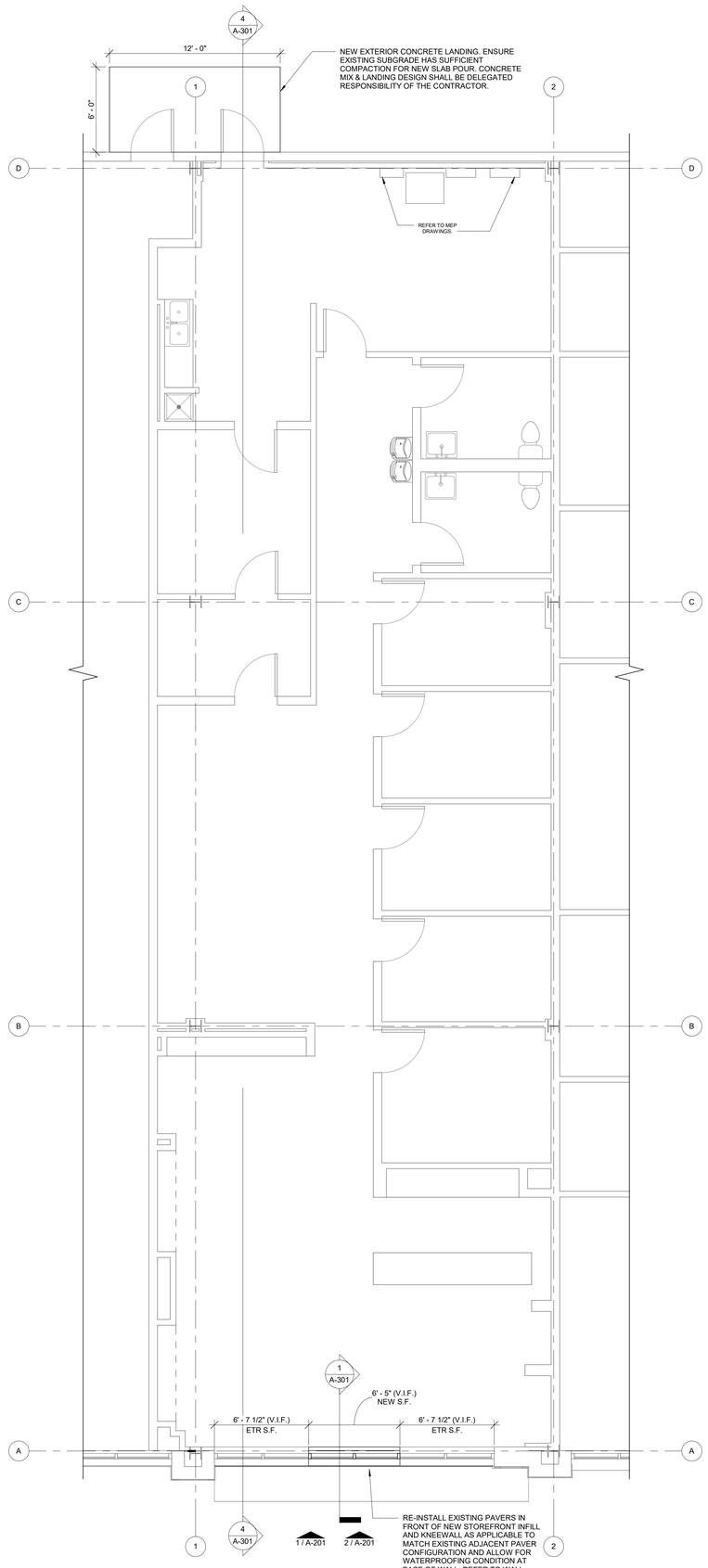
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ALICIA SOLBERG
No. 043558
FRANCE
USA

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DOCUMENT DATE:	07/02/2026

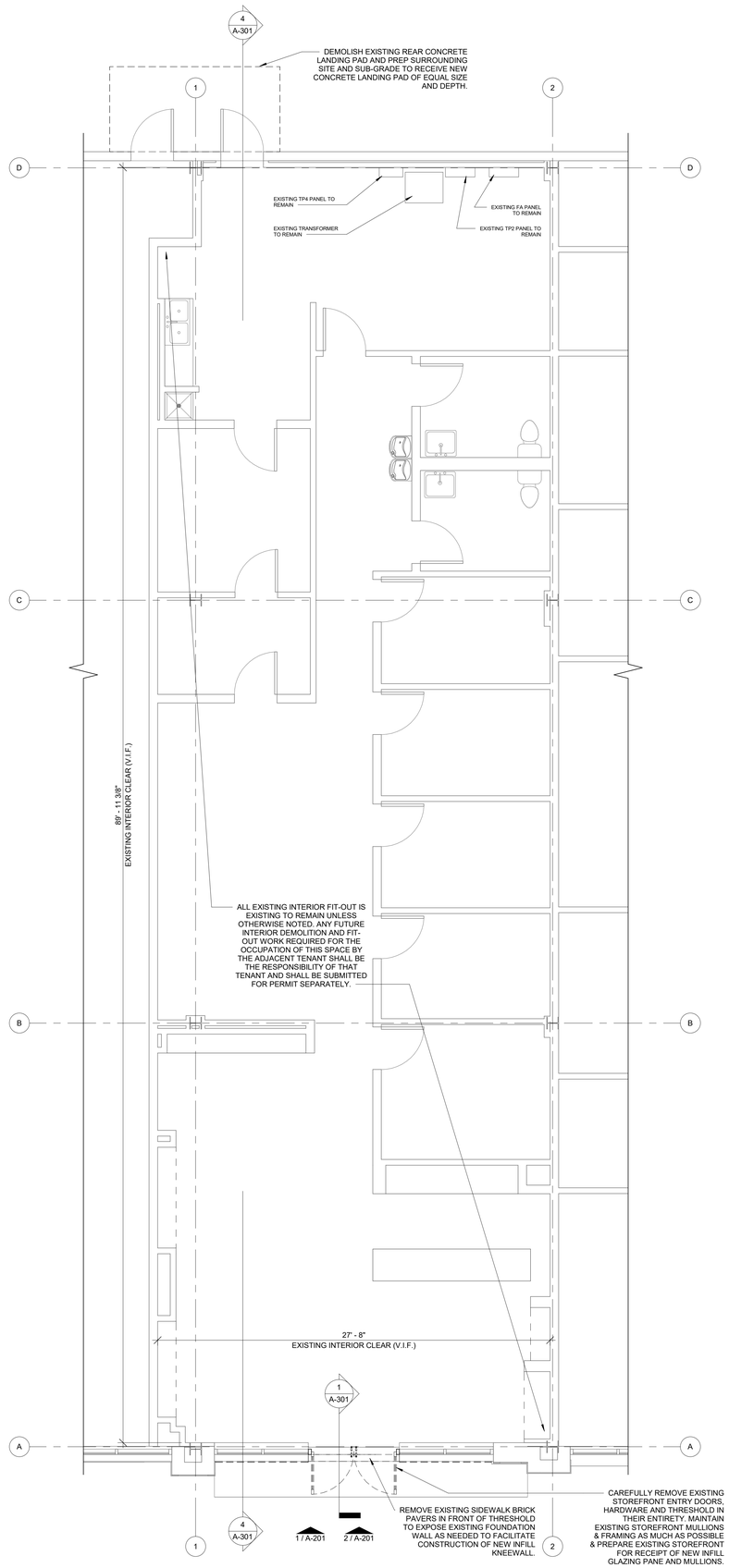
LIFE SAFETY PLAN

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2 NEW WORK FLOOR PLAN
A-101 1/4" = 1'-0"



1 DEMOLITION FLOOR PLAN
A-101 1/4" = 1'-0"

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Canton, MA 02021

JOB NUMBER: 2026021

ISSUE BLOCK		
1	Permit	07/02/26

Seal:

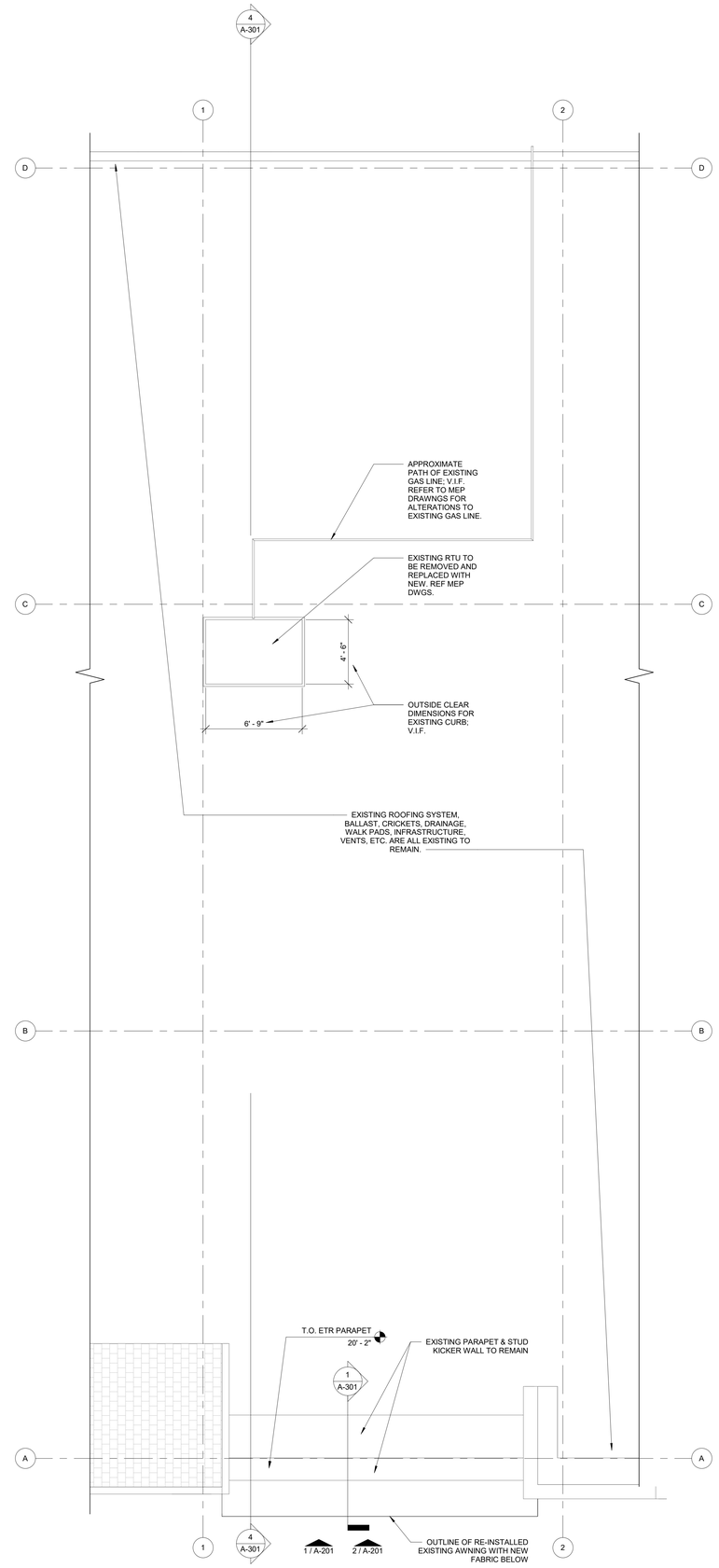
REGISTERED ARCHITECT
ANGEL SOLORZANO
No. 047006
FRAMINGHAM, MASS.

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DOCUMENT DATE:	07/02/2026

FLOOR PLANS

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A-101

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4-A-101-FLOOR PLANS-Plotted



1 A-121 NEW WORK ROOF PLAN
1/4" = 1'-0"

N

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1A-121-ROOF PLAN Permit

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Needham, MA, 02492

t: (617) 905-4043

Client:

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Suite #100
Newton, MA, 02462

t: (516) 869-2951

Consultant:

Village Shoppes Suite 13

95 Washington Street, #222
Canton, MA 02021

JOB NUMBER: 2026021

ISSUE BLOCK		
1	Permit	07/02/26

Seal:



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DOCUMENT DATE:	07/02/2026

ROOF PLAN

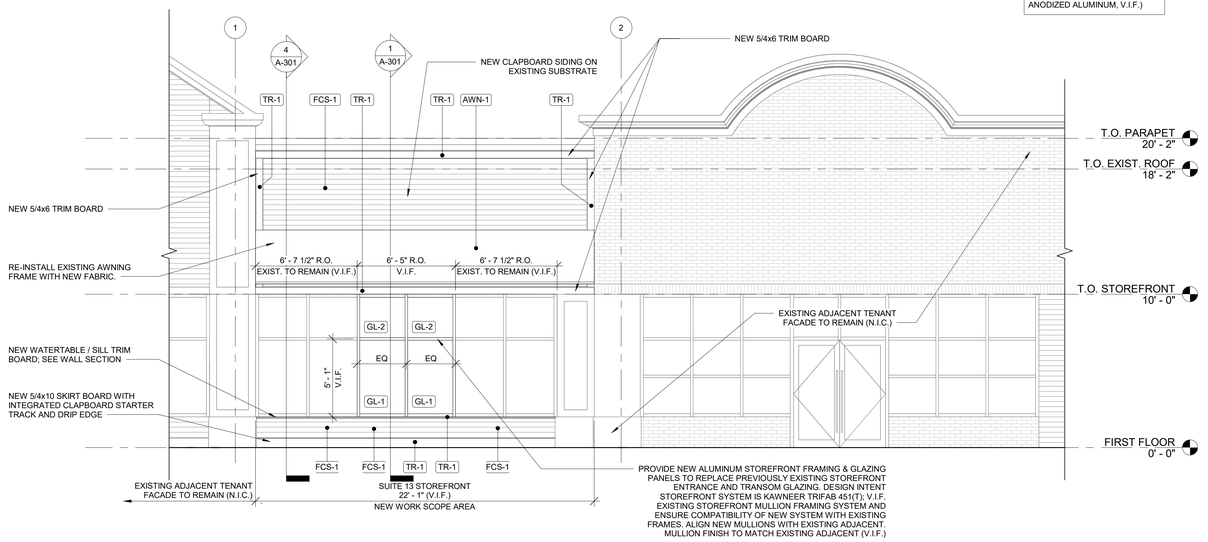
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A-121

A201_EXTERIOR FINISH LEGEND

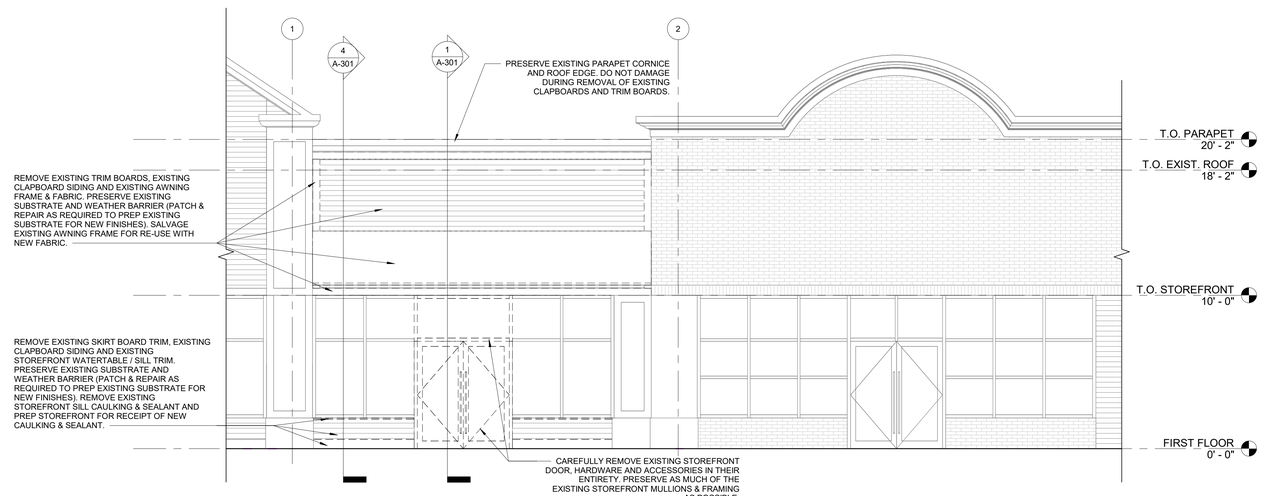
Symbol	Manufacturer	Series	Color/Pattern	Comments
TR-1	AZEK (or approved equal)	CLASSIC TRIM	WHITE	PROVIDE TRIM BOARDS, SKIRT BOARDS AND CORNERBOARDS AS NOTED ON DRAWINGS.
FCS-1	JAMES HARDIE (or approved equal)	HARDIE PLANK (SMOOTH FINISH)	COBBLE STONE	CLAPBOARD INSTALLATION WITH 4" PLANK WIDTH EXPOSED TO WEATHER. FOLLOW MANUFACTURER'S INSTALLATION GUIDELINES.
AWN-1	SUNBRELLA (or approved equal)	SUNBRELLA SHADE	BLACK	NEW FABRIC TO REPLACE EXISTING FABRIC. RE-USE EXISTING AWNING FRAMES (V.I.F.). CONFIRM FINAL FABRIC COLOR WITH OWNER, TENANT AND ARCHITECT PRIOR TO PURCHASING.

GLAZING SCHEDULE	
GL-1	1" INSULATED LOW-E, TEMPERED GLAZING
GL-2	1" INSULATED LOW-E, NON-TEMPERED GLAZING
GL-3	1/4" CLEAR, TEMPERED GLAZING

STOREFRONT FINISH
ALL NEW ALUMINUM STOREFRONTS SHALL MATCH THE FINISH OF EXISTING STOREFRONT (ASSUMED: ANODIZED ALUMINUM, V.I.F.)



2 A-201 SOUTH ELEVATION - NEW
1/4" = 1'-0"



1 A-201 SOUTH ELEVATION (DEMO)
1/4" = 1'-0"

sparc

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43 Valley Road
Needham, MA, 02492

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Consultant:

Village Shoppes Suite 13
95 Washington Street, #222
Canton, MA 02021
JOB NUMBER: 2026021

ISSUE BLOCK		
1	Permit	07/02/26

Seal:

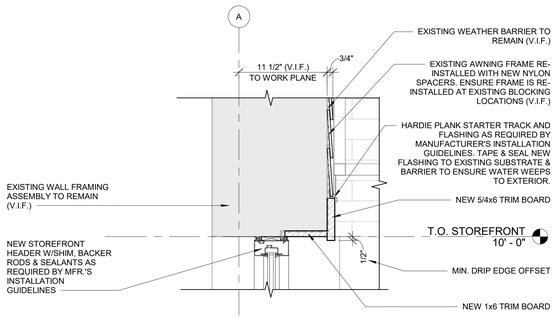


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DOCUMENT DATE:	07/02/2026

ELEVATIONS

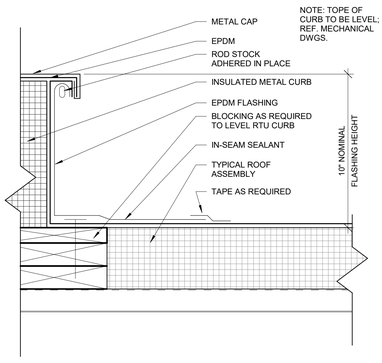
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A4-ZOY_ELEVATIONS.rvt Permit



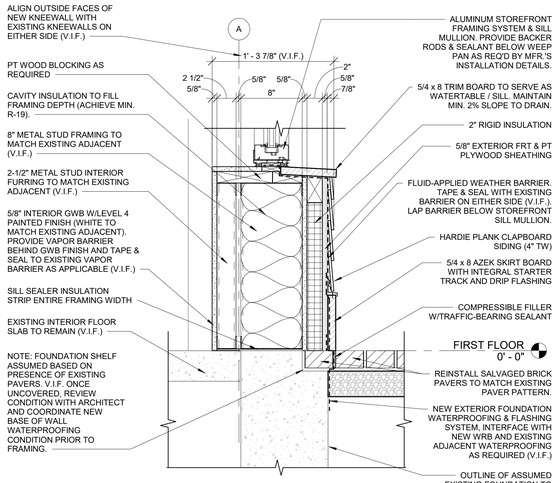
3 A-301 **Section Detail @ New Storefront Header**

1 1/2" = 1'-0"



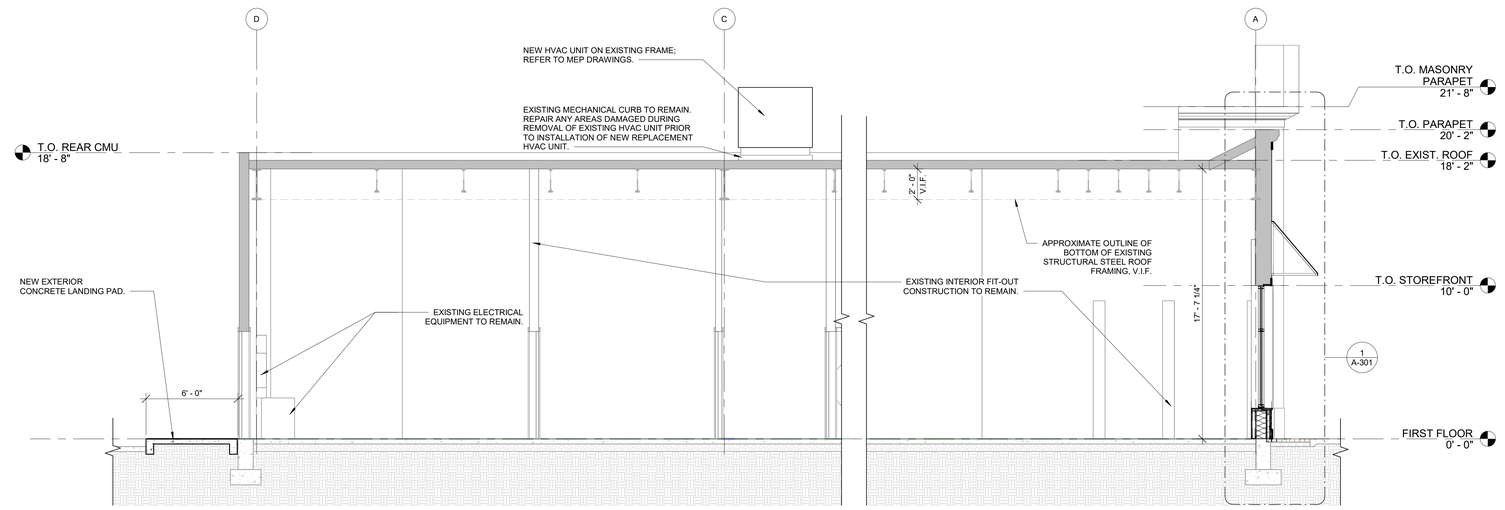
5 A-301 **MECHANICAL CURB DETAIL**

3" = 1'-0"



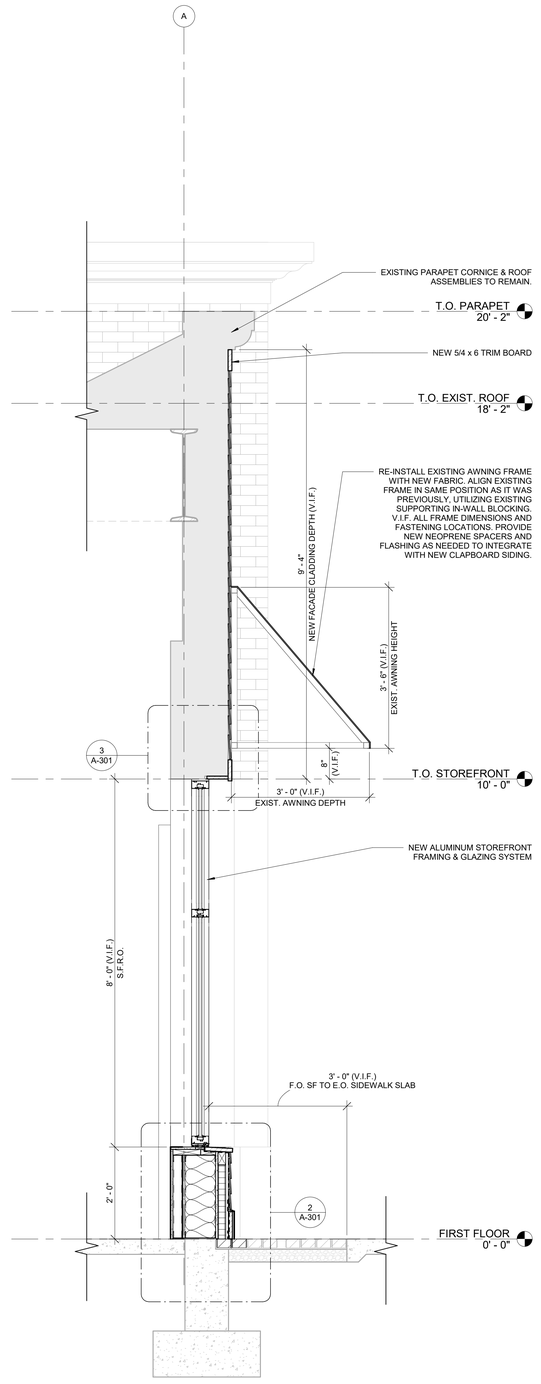
2 A-301 **Section Detail @ New Kneewall Base**

1 1/2" = 1'-0"



4 A-301 **Longitudinal Building Section**

1/4" = 1'-0"



1 A-301 **Wall Section @ Tenant Storefront**

3/4" = 1'-0"

sparc

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t: (516) 869-2951

Consultant:

Village Shoppes Suite 13

95 Washington Street, #222
Canton, MA 02021

JOB NUMBER: 2026021

ISSUE BLOCK		
1	Permit	07/02/26

Seal:

REGISTERED ARCHITECT
ANGEL SOLDEVILA
No. 047006
FRANCE
MASSACHUSETTS

CHECKED BY:	AS
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DOCUMENT DATE:	07/02/2026

SECTIONS

SHEET:
A-301

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4-A-301-SECTIONS-Permit

LEGEND NOTES

MOUNTING HEIGHTS SHALL BE AS INDICATED UNLESS INDICATED OTHERWISE ON ELECTRICAL DRAWINGS OR ARCHITECTURAL ELEVATIONS

ALL SYMBOLS MAY NOT BE SHOWN ON PLANS

HOMERUNS

P-1 — HOMERUN TO PANELBOARD. "P" DENOTES PANEL, "1" DENOTES CIRCUIT NUMBER, 20 AMP 1 POLE CIRCUIT BREAKER UNLESS INDICATED OTHERWISE. WIRING SHALL BE 2#12+1#12G IN 3/4"C AT MINIMUM.

30/3 → P-2,4,6 (2#10+1#10G IN 3/4"C.) — MULTI-POLE HOMERUN TO PANELBOARD. "P" DENOTES PANEL, "2,4,6" DENOTES CIRCUIT NUMBERS, "30/3" DENOTES 30 AMP 3 POLE CIRCUIT BREAKER. WIRING SHALL BE AS INDICATED.

RACEWAYS AND WIRING

- E — 2#10,#10G EMERGENCY ONLY WIRING
- CT — CABLE TRAY - REFER TO SPECIFICATIONS FOR REQUIREMENTS
- FLEXIBLE CONNECTION TO EQUIPMENT. RACEWAY AND CONDUCTOR RATING TO MATCH ASSOCIATED BRANCH CIRCUIT OR FEEDER.
- BRANCH CIRCUIT OR FEEDER CONCEALED IN FINISHED AREA.
- BRANCH CIRCUIT OR FEEDER CONCEALED UNDER FINISHED FLOOR.
- BRANCH CIRCUIT OR FEEDER TURNING UP TOWARDS OBSERVER.
- BRANCH CIRCUIT OR FEEDER TURNING DOWN AWAY FROM OBSERVER.
- CONDUIT STUBBED ABOVE CEILING.

LIGHTING FIXTURES

- A / 2a — SURFACE OR RECESSED MOUNTED LIGHTING FIXTURE ON NORMAL CIRCUIT, "A" DENOTES FIXTURE TYPE, "2" DENOTES CIRCUIT NUMBER, "a" DENOTES SWITCH CONTROL.
- LIGHTING FIXTURE WIRED TO CONSTANT-ON OR NORMAL EMERGENCY CIRCUIT
- SURFACE OR RECESSED MOUNTED LINEAR LIGHTING FIXTURE
- PENDANT MOUNTED LINEAR LIGHTING FIXTURE
- ROUND RECESSED LIGHTING FIXTURE
- DECORATIVE PENDANT LIGHTING FIXTURE
- WALL MOUNTED LIGHTING FIXTURE
- WALL WASH OR DIRECTIONAL LIGHTING FIXTURE
- CEILING MOUNTED ILLUMINATED EXIT SIGN. SINGLE OR DOUBLE FACE, WITH OR WITHOUT ARROWS AS INDICATED ON DRAWINGS
- WALL MOUNTED ILLUMINATED EXIT SIGN - SHADING INDICATES FACE PLATE(S)
- SELF-CONTAINED EMERGENCY LIGHTING UNIT
- SIZ — REMOTE EMERGENCY LIGHTING HEADS - SINGLE OR DOUBLE AS SHOWN.

LIGHTING CONTROL DEVICES

LIGHTING CONTROL SYSTEM:
REFER TO LIGHTING CONTROL SYSTEM RISER, DETAILS, & NOTES FOR ADDITIONAL REQUIREMENTS AND DEVICES.

- Sa — SINGLE POLE TOGGLE SWITCH; SUBSCRIPT INDICATES LIGHTING FIXTURE CONTROL.
- S3a — THREE WAY TOGGLE SWITCH; SUBSCRIPT INDICATES LIGHTING FIXTURE CONTROL..
- S4a — FOUR WAY TOGGLE SWITCH; SUBSCRIPT INDICATES LIGHTING FIXTURE CONTROL.

RECEPTACLES AND POWER DEVICES

- IG 48" 2 WP — DUPLEX RECEPTACLE, "2" DENOTES CIRCUIT NUMBER, "48" DENOTES MOUNTING HEIGHT (18" UNLESS OTHERWISE NOTED), "GFI" DENOTES GROUND FAULT CIRCUIT INTERRUPTER TYPE DEVICE, "IG" DENOTES ISOLATED GROUND TYPE DEVICE, "WP" DENOTES WEATHER PROOF COVER
- DUPLEX RECEPTACLE MOUNTED 6" ABOVE COUNTER TOP OR AS INDICATED ON ARCHITECTURAL PLANS
- X — DOUBLE DUPLEX RECEPTACLE;"X" INDICATES 1 RECEPTACLE CONTROLLED BY ROOM PLUG LOAD CONTROLLER. CONTROLLED RECEPTACLE SHALL BE EQUIPPED WITH PERMANENT MARKING.
- DOUBLE DUPLEX RECEPTACLE MOUNTED 6" ABOVE COUNTER TOP OR AS INDICATED ON THE ARCHITECTURAL PLANS
- X — DUPLEX RECEPTACLE;"X" INDICATES 1/2 RECEPTACLE CONTROLLED BY ROOM PLUG LOAD CONTROLLER. CONTROLLED RECEPTACLE SHALL BE EQUIPPED WITH PERMANENT MARKING.
- SINGLE RECEPTACLE
- DUPLEX GROUND FAULT CIRCUIT INTERRUPTER RECEPTACLE
- DUPLEX GROUND FAULT CIRCUIT INTERRUPTER RECEPTACLE MOUNTED 6" ABOVE COUNTER TOP OR AS INDICATED ON ARCHITECTURAL PLANS
- DOUBLE DUPLEX GROUND FAULT CIRCUIT INTERRUPTER RECEPTACLE MOUNTED 6" ABOVE COUNTER TOP OR AS INDICATED ON THE ARCHITECTURAL PLANS
- CEILING MOUNTED DUPLEX RECEPTACLE

POWER DISTRIBUTION SYSTEM

- DISTRIBUTION PANEL
- PANELBOARD, SURFACE MOUNTED
- PANELBOARD, FLUSH MOUNTED
- J — JUNCTION BOX, SIZED PER NEC
- 2 — MOTOR, "2" DENOTES HORSEPOWER
- MAGNETIC MOTOR STARTER WITH ENCLOSURE, MINIMUM SIZE NEMA 1
- 30/3 — NON-FUSED DISCONNECT SWITCH: "30/3" DENOTES 30 AMP/3 POLE SWITCH
- 30/20/3 — FUSED DISCONNECT SWITCH: "30/20/3" DENOTES 30 AMP/3 POLE SWITCH, 20 AMP FUSES
- COMBINATION MAGNETIC STARTER AND FUSED DISCONNECT SWITCH. SIZE OF STARTER, SWITCH AND FUSE AS REQUIRED
- T15 — DRY-TYPE DISTRIBUTION TRANSFORMER. "15" DENOTES SIZE.
- GROUND

ABBREVIATIONS

3R	NEMA 3R RATING	JB	JUNCTION BOX
4X	NEMA 4X RATING	KCMIL	ONE THOUSAND CIRCULAR MILS
A/AMP	AMPERES	KVA	KILOVOLT-AMPERES
ADA	AMERICAN WITH DISABILITIES ACT	KW	KILOWATTS
AFF	ABOVE FINISHED FLOOR	MCA	MINIMUM CIRCUIT AMPS
AHJ	AUTHORITY HAVING JURISDICTION	MCB	MAIN CIRCUIT BREAKER
AIC	AMPERE INTERRUPTING CAPACITY	MD	MOTORIZED DAMPER
ATS	AUTOMATIC TRANSFER SWITCH	MLO	MAIN LUGS ONLY
AWG	AMERICAN WIRE GAUGE	MOCP	MAXIMUM OVER-CURRENT PROTECTION
C	CONDUIT	N	NEUTRAL
CB	CIRCUIT BREAKER	NEC	NATIONAL ELECTRICAL CODE
CD	CANDELA	NIC	NOT IN CONTRACT
DWG	DRAWING	NTS	NOT TO SCALE
E	WIRED ON EMERGENCY CIRCUIT	Ø	PHASE
EC	ELECTRICAL CONTRACTOR	P	POLE
FLA	FULL LOAD AMPERES	PVC	POLYVINYL CHLORIDE
G	GROUND	T/D	TEL/DATA
GC	GENERAL CONTRACTOR	TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSOR
GFCI	GROUND FAULT CIRCUIT INTERRUPTER	TYP	TYPICAL
HP	HORSE POWER	UL	UNDERWRITERS LABORATORIES
		UNO	UNLESS NOTED OTHERWISE
		V	VOLTS
		VA	VOLT-AMPERE
		W	WATT
		WP	WEATHERPROOF

DEMOLITION AND REMOVAL WORK

1. REMOVE ALL ELECTRICAL EQUIPMENT, WIRING, AND OTHER ELECTRICAL WORK AS REQUIRED. DISCONNECT LOAD AND LINE END OF CONDUCTORS FEEDING DEVICES WHICH ARE TO BE REMOVED OR ABANDONED, REMOVE CONDUCTORS NO LONGER IN USE. CUT BACK TO FLOOR, WALL, OR CEILING AND PLUG BOTH ENDS OF CONCEALED CONDUITS MADE OBSOLETE BY THIS ALTERATION. REMOVE EXPOSED OR ABANDONED CIRCUITS AND OUTLETS; REMOVE MATERIAL AND EQUIPMENT AND DISPOSE OF AS DIRECTED.
2. WHEREVER IT IS REQUIRED TO DISCONNECT OR REMOVE ANY PART OF AN EXISTING CIRCUIT, IMMEDIATELY RECONNECT THAT CIRCUIT OR REESTABLISH SERVICE IN THE REMAINING PORTION OF THE CIRCUIT.* THE WORK SHALL ALSO INCLUDE THE REMOVAL OF MATERIALS AS DIRECTED. PRIOR TO REMOVING EQUIPMENT AND MATERIAL FROM PROJECT SITE, THE BUILDING MANAGER OR OWNER WILL INSPECT AND ADVISE WHICH ITEMS WILL BE STORED.
3. WHERE EXISTING RECEPTACLES AND/OR SWITCHES ARE LOCATED IN COLUMNS AND/OR EXTERIOR WALLS, AND ARE NOT TO BE REUSED, REMOVE RECEPTACLE AND CAP OUTLET BOX. RECEPTACLES SHOWN ON PARTITIONS TO BE REMOVED SHALL HAVE ALL WIRING AND CONDUIT REMOVED AS WELL.
4. WHERE PRESENT WORK IS DAMAGED IN THE EXECUTION OF THIS CONTRACT, OR WHERE OPENINGS ARE LEFT DUE TO THE REMOVAL OF CONDUITS, EQUIPMENT, OR APPARATUS, THE SAME SHALL BE REPAIRED OR CLOSED UP TO CORRESPOND IN MATERIAL, QUALITY, SHAPE, AND FINISH WITH THAT OR SIMILAR AND ADJOINING WORK, UNLESS OTHERWISE CALLED FOR.
5. SHOULD ANY DAMAGE DUE TO THE EXECUTION OF THIS CONTRACT OCCUR TO THE FURNITURE, FIXTURES, OR ANY OTHER EQUIPMENT OR APPARATUS, SUCH DAMAGES SHALL BE PROPERLY REPAIRED WITH THE SUPPLY OF NEW ARTICLES AND MADE GOOD WITHOUT EXTRA CHARGE.
6. WHERE REMOVAL OF EXISTING ELECTRICAL EQUIPMENT WILL RESULT IN OUTAGES IN AREA NOT TO BE DEMOLISHED, THIS CONTRACTOR SHALL COORDINATE IN ADVANCE AND OBTAIN THE APPROVAL OF THE BUILDING MANAGER OR OWNER.

LIGHTING FIXTURE NOTES

1. PROVIDE ACCESSORIES AND MOUNTING HARDWARE FOR ALL FIXTURES.
2. COLORS SHALL BE AS SELECTED BY ARCHITECT.
3. COORDINATE EXACT LOCATIONS OF ALL FIXTURES WITH ARCHITECT'S REFLECTED CEILING PLAN, ELEVATIONS, SECTIONS, AND THE WORK OF OTHER TRADES PRIOR TO ROUGH-IN.
4. E.C. SHALL ENSURE THAT ALL PROPOSED SWITCHES AND DIMMER SWITCHES ARE COMPATIBLE WITH THE LIGHT FIXTURE(S) INDICATED TO BE CONTROLLED. INSTALL ALL SWITCHES AND DIMMER SWITCHES PER MANUFACTURER'S RECOMMENDATIONS AND REQUIREMENTS.
5. ALL SELF CONTAINED EMERGENCY LIGHTING UNITS AND EXIT LIGHTING IN THE BUILDING SHALL BE CONNECTED TO THE NEAREST UN-SWITCHED LIGHTING CIRCUIT SERVING THE AREA WITH 2#12 & 1#12G, 3/4" CONDUIT UNLESS OTHERWISE NOTED.
6. LOCATIONS OF ALL SWITCHES SHALL COMPLY WITH ADA CRITERIA.
7. WHERE SWITCH CONTROLS ("a", "b", ETC.) ARE INDICATED, WIRE THE SWITCHES TO THE RESPECTIVE LIGHT FIXTURE. IF A FIXTURE HAS TWO OR MORE SWITCH DESIGNATIONS, WIRE FIXTURE SO THAT IT WILL BE CONTROLLED BY THE SWITCHES INDICATED.
8. WIRE EXIT SIGNS TO LIGHTING CIRCUIT SERVING THE AREA AHEAD OF ALL CONTROLS.
9. WIRE NIGHT LIGHTING FIXTURES FOR 24/7 OPERATION VIA UN-SWITCHED CIRCUIT AS INDICATED.
10. METAL ROOF DECKS SHALL NOT BE TAPPED FOR SUPPORT OF ANY LIGHTING FIXTURES OR ELECTRICAL EQUIPMENT. PROVIDE UNISTRUT OR OTHER SUPPLEMENTAL SUPPORT FITTINGS TO BE ATTACHED TO BUILDINGS STRUCTURAL FRAMING AS REQUIRED TO SUPPORT ALL LIGHTING FIXTURES AND ELECTRICAL EQUIPMENT.

DEMOLITION LEGEND

- ETR — "ETR" DENOTES EXISTING ELECTRICAL DEVICE WHICH IS TO REMAIN.
- ETD — "ETD" DENOTES EXISTING ELECTRICAL DEVICE WHICH IS TO BE DEMOLISHED. PULL BACK WIRING AND CONDUIT BACK TO NEXT ACTIVE OUTLET OR POWER SOURCE.
- ETRL — "ETRL" DENOTES EXISTING ELECTRICAL DEVICE TO BE DISCONNECTED AND RELOCATED. EXISTING CIRCUIT SHALL BE EXTENDED AS REQUIRED TO NEW LOCATION OF EXISTING ELECTRICAL DEVICE.
- RL — "RL" DENOTES NEW LOCATION OF RELOCATED EXISTING ELECTRICAL DEVICE.
- ETRP — "ETRP" DENOTES EXISTING DEVICE TO BE REMOVED AND REPLACED. EXISTING CIRCUIT/WIRING AND BACK BOX SHALL REMAIN. NEW DEVICE SHALL BE LOCATED IN PLACE.
- RP — "RP" DENOTES REPLACED DEVICE.

BRANCH CIRCUIT WIRING NOTES

1. WIRING IS SHOWN ON DRAWINGS ONLY FOR SPECIFIC ROUTES OR SPECIAL CONDITIONS.
2. WIRING AND CONDUIT SHALL BE REQUIRED BETWEEN ALL OUTLETS INDICATED WITH CIRCUIT NUMBERS AND PANEL DESIGNATIONS.
3. ALL SWITCH CONTROLS SHALL BE PROVIDED WITH WIRING AND CONDUIT AS REQUIRED.
4. ALTHOUGH ALL BRANCH CIRCUIT WIRING AND CONDUIT IS NOT SHOWN, IT IS THE INTENT OF THESE DOCUMENTS THAT A COMPLETE BRANCH CIRCUIT WIRING SYSTEM BE INSTALLED.
5. A GREEN GROUNDING CONDUCTOR SHALL BE RUN WITH ALL CIRCUITS. VERIFY CONDUIT SIZE TO ENSURE IT CAN ACCOMMODATE ALL PHASE, NEUTRAL AND GROUND CONDUCTORS.
6. PROVIDE A NEUTRAL CONDUCTOR TO ALL NEW LIGHTING SWITCH BOXES PER NEC ARTICLE 404.2.
7. IN ALL NON-DWELLING TYPE OCCUPANCIES, ALL 125-VOLT THROUGH 250-VOLT RECEPTACLES SUPPLIED BY SINGLE-PHASE BRANCH CIRCUITS RATED 150 VOLTS OR LESS TO GROUND, 50 AMPERES OR LESS, AND ALL RECEPTACLES SUPPLIED BY THREE-PHASE BRANCH CIRCUITS RATED 150 VOLTS OR LESS TO GROUND, 100 AMPERES OR LESS, SHALL HAVE GROUND-FAULT CIRCUIT-INTERRUPTER PROTECTION FOR PERSONNEL PER NEC ARTICLE 210.8(B)(2).
8. WHERE EXISTING SWITCHES AND RECEPTACLES ARE INDICATED TO REMAIN, THIS CONTRACTOR SHALL REPLACE SAID DEVICE(S) AND DEVICE PLATE(S) WITH NEW TO MATCH THE NEW CONSTRUCTION. WHERE THEY ARE INDICATED AS RELOCATED, EXTEND BRANCH CIRCUIT WIRING TO NEW LOCATION AND PROVIDE NEW DEVICE AND DEVICE PLATE TO MATCH NEW CONSTRUCTION.

MECHANICAL/PLUMBING EQUIPMENT TAG

RTU 1 — MECHANICAL/PLUMBING EQUIPMENT TAG. "RTU" DENOTES EQUIPMENT TYPE, "1" DENOTES EQUIPMENT NUMBER. REFER TO "SCHEDULE FOR MECHANICAL/PLUMBING EQUIPMENT" FOR ALL CIRCUIT INFORMATION, INCLUDING BUT NOT LIMITED TO BRANCH CIRCUIT WIRING, CONDUIT SIZE, VOLTAGE, PHASE, DISCONNECT SWITCH, AND CIRCUIT BREAKER. REFER TO MECHANICAL, PLUMBING, AND FIRE PROTECTION PLANS FOR EXACT EQUIPMENT LOCATIONS AND EQUIPMENT TYPE ABBREVIATIONS.

FIRE ALARM SYSTEM

FIRE ALARM SYSTEM:
ALL NOTIFICATION DEVICES SHALL BE MOUNTED 80" AFF, UNLESS OTHERWISE NOTED. THE FOLLOWING DESIGNATIONS SHALL APPLY TO ALL FIRE ALARM DEVICES:

AC = ABOVE CEILING
C = CEILING MOUNTED
LF = LOW FREQUENCY
WG = WIRE GUARD
WP = WEATHERPROOF

- F — MANUAL PULL STATION; MOUNTED 48" AFF
- FAA — FIRE ALARM ANNUNCIATOR
- BATT — BATTERY CABINET
- FACP — FIRE ALARM CONTROL PANEL
- NAC — NOTIFICATION APPLIANCE CIRCUIT POWER SUPPLY SIZE PER MANUFACTURERS SYSTEM CALCULATIONS.
- S — SMOKE DETECTOR/SENSOR - BASIC SHAPE ORIENTATION NOT TO BE CHANGED.
- S — DUCT MOUNTED SMOKE DETECTOR, INSTALLED BY MECHANICAL CONTRACTOR, WIRED AND FURNISHED BY ELECTRICAL CONTRACTOR.
- 15cd — VISIBLE ONLY (STROBE) - CEILING MOUNT; CD=CANDELA RATING/SETTING
- 15cd — VISIBLE ONLY (STROBE) - WALL MOUNT; CD=CANDELA RATING/SETTING
- 15cd — COMBINATION HORN/VISIBLE; CD= CANDELA RATING/SETTING
- RI — REMOTE ALARM INDICATOR; CEILING MOUNT
- RI — REMOTE ALARM INDICATOR; WALL MOUNT
- RTS — REMOTE ALARM INDICATING AND TEST SWITCH; MOUNTED 7'-0" AFF.
- AOM — ADDRESSABLE OUTPUT CONTROL MODULE
- AIM — ADDRESSABLE INPUT MONITOR MODULE
- AI — ADDRESSABLE ISOLATION MODULE
- CR — ADDRESSABLE CONTROL RELAY
- WFT — FLOW SWITCH (WATER), FURNISHED AND INSTALLED BY FIRE PROTECTION SUBCONTRACTOR, WIRED BY THE ELECTRICAL SUBCONTRACTOR.
- VS — TAMPER SWITCH, FURNISHED AND INSTALLED BY FIRE PROTECTION SUBCONTRACTOR, WIRED BY THE ELECTRICAL SUBCONTRACTOR.

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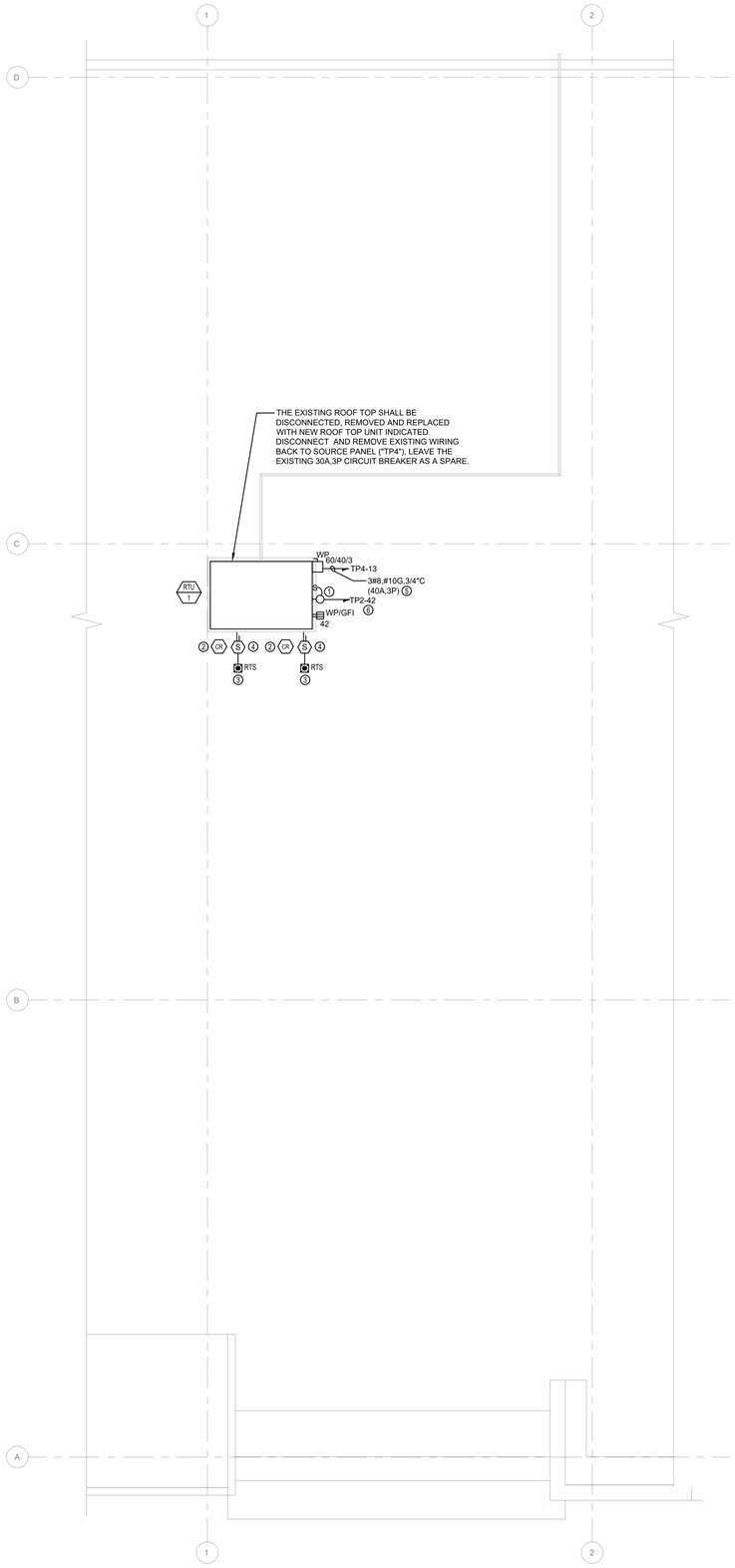
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ELECTRICAL LEGEND, NOTES, & FIXTURE SCHEDULES

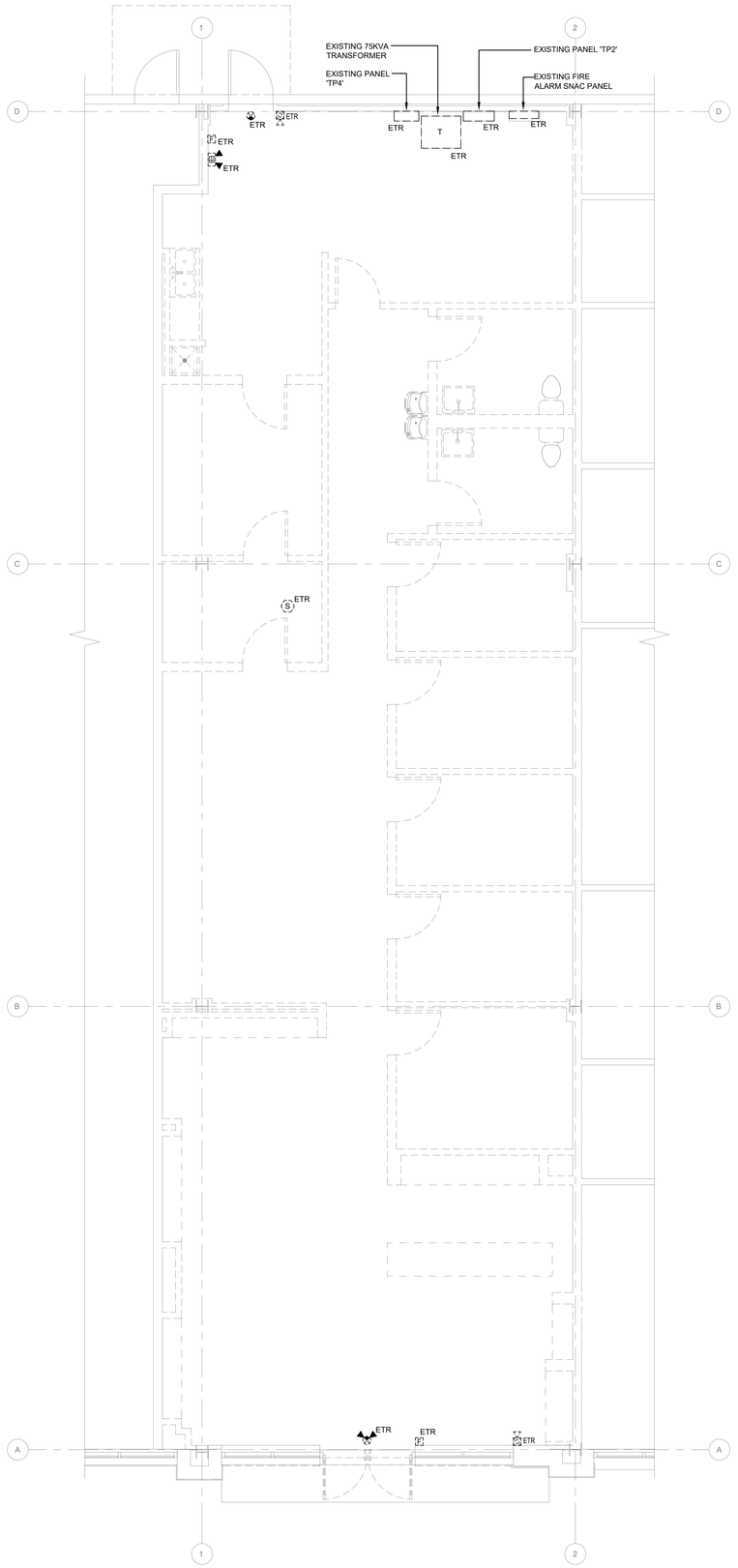
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2 E-100 ELECTRICAL ROOF PLAN

1/4" = 1'-0"



1 E-100 ELECTRICAL FLOOR PLAN

1/4" = 1'-0"

KEYED NOTES:

1. PROVIDE UTILITY LIGHT FIXTURE AT INDICATED ROOF MOUNTED MECHANICAL UNIT EQUAL TO E-CONOLITE #E-VTBO02A-WM401G.
2. PROVIDE ADDRESSABLE CONTROL RELAY/MODULE FOR SHUT DOWN OF MECHANICAL ROOF TOP UNIT ALARM CONDITION.
3. COORDINATE MOUNTING LOCATIONS OF DUCT SMOKE DETECTOR REMOTE TEST STATIONS WITH THE LOCAL FIRE DEPARTMENT.
4. COORDINATE THE INSTALLATION OF ALL DUCT MOUNTED SMOKE DETECTORS WITH THE MECHANICAL CONTRACTOR.
5. PROVIDE NEW 40A,3P CIRCUIT BREAKER WITHIN EXISTING PANELBOARD "TP4" TO SERVE NEW MECHANICAL ROOF TOP UNIT. NEW CIRCUIT BREAKER SHALL MATCH THE EXISTING PANELBOARD MANUFACTURER AND AIC RATING AND SHALL BE INSTALLED WITHIN EXISTING PANELBOARD SPACES.
6. UTILIZE EXISTING 20A, 1P CIRCUIT BREAKER WITH EXISTING PANELBOARD "TP2" (MADE AVAILABLE BY DEMOLITION) FOR NEW BRANCH CIRCUITRY TO RECEPTACLE AND UTILITY LIGHT FIXTURE LOCATED AT NEW ROOF TOP MECHANICAL UNIT.

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ELECTRICAL FLOOR AND ROOF PLANS

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PART 1 - GENERAL

1.01 GENERAL REQUIREMENTS

A. THESE SPECIFICATIONS AND DRAWINGS ARE INTENDED TO SERVE JOINTLY AS A BASIS UPON WHICH THE CONTRACTOR SHALL SUBMIT A PRICE FOR MATERIAL AND LABOR PROVISIONS.

B. IT IS NOT INTENDED THAT THE PLANS OR SPECIFICATION SHOW OR STATE EVERY DETAILED REQUIREMENT OF THE WORK, BUT RATHER THAT THEY FURNISH ADEQUATE INFORMATION FOR AN EXPERIENCED CONTRACTOR TO PROVIDE A COMPLETELY ACCEPTABLE INSTALLATION. THE GENERAL CONDITIONS FORM A PART OF THESE SPECIFICATIONS WHETHER ATTACHED HERETO OR NOT, SHALL BE CAREFULLY EXAMINED BEFORE SUBMITTING A PROPOSAL. WHERE GENERAL CONDITIONS CLAUSES ARE REPEATED IN THIS SECTION, IT SHALL BE UNDERSTOOD AS CALLING SPECIAL ATTENTION TO THEM, OR AS A FURTHER QUALIFICATION, AND SHALL NOT BE ASSUMED AS OMITTING ANY OTHER CLAUSES. NO GENERAL CONDITIONS REFERRING TO THE WORK INCLUDED HEREIN SHALL BE CONSIDERED AS WAIVED UNLESS SPECIFICALLY STATED HEREIN.

C. BEFORE SUBMITTING PROPOSAL, EXAMINE ALL PLANS RELATING TO THIS WORK, VERIFY ALL GOVERNING CONDITIONS AT THE SITE, BECOME FULLY INFORMED AS TO THE EXTENT AND CHARACTER OF THE WORK REQUIRED AND ITS RELATION TO THE WORK OF OTHER TRADES. SUBMISSION OF A COST PROPOSAL (BID) WILL BE JUDGED AS EVIDENCE THAT THE SITE EXAMINATION HAS BEEN MADE. NO CONSIDERATION WILL BE GRANTED FOR ANY ALLEGED MISUNDERSTANDING OF THE MATERIALS TO BE FURNISHED FOR WORK TO BE DONE. IT BEING UNDERSTOOD THAT THE SUBMISSION OF A PROPOSAL IS AN AGREEMENT TO ALL CONDITIONS REFERRED TO HEREIN OR INDICATED ON THE PLANS.

D. PROPOSAL MUST INCLUDE EVERYTHING REQUIRED TO PROVIDE A COMPLETE INSTALLATION AS CONTEMPLATED IN A SPECIFICATIONS AND PLANS, WHETHER SPECIFICALLY SHOWN OR SPECIFIED OR NOT: INCLUDED ARE LABOR, MATERIALS, EQUIPMENT, LIGHTS, TOOLS, SCAFFOLDING, ETC. NECESSARY TO COMPLETE INSTALLATION OF EVERYTHING DESCRIBED, SHOWN OR REASONABLY IMPLIED.

E. ANY DISCREPANCIES BETWEEN THESE SPECIFICATIONS AND THE ACCOMPANYING PLANS, OR THESE SPECIFICATIONS AND PLANS AND THE SPECIFICATIONS OF OTHER TRADES, SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT PRIOR TO THE SUBMISSION OF THE BID. FAILURE TO COMPLY WITH THE ABOVE SHALL ALLOW THE ARCHITECT TO MAKE A FINAL AND BINDING DECISION AT A LATER DATE AND NO ALLOWANCE WILL BE GIVEN IF THE MORE EXPENSIVE OF THE ITEM IN QUESTION IS SELECTED.

F. THE WORK CALLED FOR IN THESE PLANS AND SPECIFICATIONS SHALL BE COORDINATED WITH THE STRUCTURE, WORK OF ALL RELATED TRADES, AND SHALL BE SO ARRANGED THAT THERE WILL BE NO DELAY IN THE PROPER INSTALLATION AND COMPLETION OF ANY PART OF EACH RESPECTIVE WORK. WHEREIN IT MAY BE INTERRELATED WITH THIS CONTRACT ALL WORK CAN PROCEED IN ITS NATURAL SEQUENCE WITHOUT UNNECESSARY DELAY, THE CONTRACTOR IS RESPONSIBLE FOR ANY ADDITIONAL COST AND DELAYS IN THE WORK RESULTING FROM SUBSTITUTION UNDER THIS DIVISION: INCLUDING, BUT NOT LIMITED TO, ANY CHANGES, INDECISION, INSTALLATION OR THE WORK OF OTHER TRADES.

G. ELECTRICAL DRAWINGS ARE DIAGRAMMATIC (EXCEPT WHERE DIMENSIONED) AND INDICATE THE GENERAL ARRANGEMENT OF SYSTEM AND WORK. FOLLOW ARCHITECTURAL, STRUCTURAL, AND MANUFACTURER'S SHOP DRAWINGS FOR GREATER ACCURACY. CONSULT ENGINEER IN CASE OF DOUBT OR CONFLICT, UNLESS, NOTED, FIXED DIMENSIONS ARE BASED ON THE PRODUCT OF ONE MANUFACTURER, VERIFY DIMENSIONS WITH THE SHOP DRAWINGS OF THE MATERIAL ACTUALLY APPROVED OR PURCHASED.

H. EXACT LOCATION OF ALL EQUIPMENT, PANELS, PULL BOXES, FEEDERS, FIXTURES, ETC., SHALL BE APPROVED BY THE ARCHITECT, AND OWNER PRIOR TO THE INSTALLATION OF THE SAME.

I. CONTRACTOR SHALL FURNISH ALL NECESSARY OUTLETS, SUPPORTS, FITTINGS, AND ACCESSORIES TO FULFILL APPLICABLE CODES, REGULATIONS, AND THE BEST PRACTICES OF THE TRADE FOR INSTALLATION OF ALL ELECTRICAL WORK.

J. EXPOSED CONDUITS CAN BE INSTALLED BUT IN NO CASE SHALL BE INSTALLED LESS THAN NINE FEET ABOVE THE FINISHED FLOOR OR AS NOTED. CONDUITS INSTALLED IN THE AREA WHERE HUNG CEILING OR OTHER FURRED SPACES ARE INDICATED SHALL BE INSTALLED CONCEALED. SHOULD ANY WORK REQUIRE SUBSEQUENT MODIFICATION OR RELOCATION TO AVOID INTERFERENCE OR CONFLICTS WITH OTHER WORK, SUCH CHANGES SHALL BE MADE WITHOUT ADDITIONAL COST TO THE OWNER.

K. ANY NECESSARY ELECTRICAL SERVICE INTERRUPTIONS SHALL BE AT A TIME CONVENIENT TO THE BUILDING OWNER.

L. ALL PENETRATIONS THROUGH SLABS AND FIRE RATED PARTITIONS SHALL BE FIRE STOPPED USING APPROVED METHOD TO MAINTAIN THE FIRE RESISTANCE RATING.

M. THE E.C. SHALL CALL FOR A FINAL PUNCH-LIST WHEN ALL ELECTRICAL WORK IS COMPLETE. IN THE EVENT REPRESENTATIVES OF THIS OFFICE ARRIVE AT THE SITE AND DEEM WORK IS NOT SUFFICIENTLY COMPLETE, NO REVIEW/PUNCH-LIST WILL OCCUR. THE E.C. WILL BE RESPONSIBLE FOR ALL COSTS ASSOCIATED WITH RESCHEDULING THE PUNCH-LIST.

1.02 SCOPE OF WORK

A. WITHOUT INTENDING TO LIMIT AND/OR RESTRICT THE SCOPE OF WORK REQUIRED AND SOLELY FOR THE CONVENIENCE OF THE CONTRACTOR, THE WORK OF THIS DIVISION SHALL, IN GENERAL COMPRISE THE FOLLOWING:

1. FURNISHED AND INSTALLING NEW LIGHT FIXTURES, LAMPS AND LIGHTING CONTROLS.
2. FURNISHING AND INSTALLING NEW FEEDERS, CONDUITS, BRANCH CIRCUIT WIRING, ETC.
3. FURNISHING AND INSTALLING NEW RACEWAYS, OUTLET BOXES, WIRING AND CONNECTIONS FOR LIGHTING FIXTURES, SWITCHES, AND RECEPTACLES.
4. PROVIDE CONNECTIONS TO ALL ELECTRICAL EQUIPMENT FURNISHED BY OTHER TRADES OR BY THE OWNER.
5. POWER WIRING FOR ALL MOTORS, INCLUDING INSTALLING ALL REQUIRED DISCONNECT SWITCHES AND MOUNTING OF STARTERS.
6. TEMPORARY LIGHT AND POWER.
7. GROUNDING.
8. CUTTING, CHANNELING, AND PATCHING.
9. FIRE ALARM SYSTEM MODIFICATIONS.
10. REMOVAL OF ELECTRICAL SYSTEM AS REQUIRED AND AS INDICATED ON PLANS

1.03 WORK NOT INCLUDED

A. FURNISHING MOTORS, MOTOR STARTER AND CONTROL DEVICES CONNECTED TO EQUIPMENT FURNISHED UNDER OTHER DIVISIONS OF THE SPECIFICATIONS. HOWEVER, ELECTRICAL CONTRACTOR WILL ERECT AND WIRE SAME, FURNISH AUXILIARY MOTOR DISCONNECTS AS REQUIRED BY DRAWINGS OR CODE.

B. FINISH PAINTING.

1.04 CODES, PERMITS, AND INSPECTIONS

A. ELECTRICAL WORK SHALL COMPLY WITH THE APPLICABLE REQUIREMENTS OF THE STATE ELECTRICAL CODE, LOCAL ORDINANCES, AND OTHER AUTHORITIES EXERCISING JURISDICTION OVER ALL ELECTRICAL CONSTRUCTION WORK AND THE PROJECT.

B. NOTHING CONTAINED IN THESE SPECIFICATIONS OR PLANS SHALL BE SO CONSTRUED AS TO CONFLICT WITH ANY LOCAL, MUNICIPAL, AND NATIONAL BOARD OF THE FIRE UNDERWRITERS REGULATIONS GOVERNING THE INSTALLATION OF WORK SPECIFIED HEREIN. ALL SUCH LAWS, ORDINANCES, AND REGULATIONS, WHERE THEY APPLY TO THIS WORK, ARE HEREBY INCORPORATED INTO AND MADE A PART OF THE SPECIFICATIONS. ALL SUCH REQUIREMENTS SHALL BE SATISFIED AT NO ADDITIONAL EXPENSE TO THE OWNER.

C. ALL REQUIRED PERMITS AND INSPECTION CERTIFICATES SHALL BE OBTAINED, PAID FOR, AND MADE AVAILABLE AT THE COMPLETION OF THE WORK.

D. COORDINATE ALL SERVICE WORK WITH THE LOCAL UTILITY COMPANIES. ALL WORK INCLUDING BUT NOT LIMITED TO, PRIMARY ELECTRIC DUCT BANKS, TRANSFORMER PADS, MANHOLES/ PULL BOXES, METERING, PROTECTION BOLLARDS AND TELEPHONE/CABLE TV SERVICE DUCT BANKS SHALL BE PROVIDED IN ACCORDANCE WITH THE LATEST REQUIREMENTS OF THE LOCAL UTILITY COMPANIES.

1.05 GUARANTEES AND CERTIFICATIONS

A. ALL WORK SHALL BE GUARANTEED TO BE FREE FROM DEFECTS, DEFECTIVE MATERIALS OR WORKMANSHIP, AS WELL AS DAMAGE TO THE WORK OF ANY/ALL TRADES RESULTING FROM THE SAME, SHALL BE REPLACED OR REPAIRED AS DIRECTED FOR DURATION OF ONE YEAR, FROM THE DATE OF ACCEPTANCE.

B. THE DATE OF ACCEPTANCE SHALL BE THE DATE OF THE FINAL PAYMENT FOR THE WORK OR THE DATE OF A FORMAL NOTICE OF ACCEPTANCE, WHATEVER IS EARLIER.

C. NON-DURABLE ITEMS, SUCH AS ELECTRIC LAMPS, SHALL BE REPLACED UP TO THE DATE OF ACCEPTANCE, SUCH THAT THEY SHALL HAVE HAD NO MORE THAN 100 HOURS USE PRIOR TO THIS DATE.

1.06 SHOP DRAWINGS AND EQUIPMENT SUBMISSIONS

A. PRIOR TO PURCHASING ANY EQUIPMENT OR MATERIAL, A LIST OF THEIR MANUFACTURERS SHALL BE SUBMITTED FOR APPROVAL.

B. PRIOR TO ASSEMBLING OR INSTALLING THE WORK, CATALOG INFORMATION AND FACTORY ASSEMBLY DRAWINGS, AS REQUIRED FOR A COMPLETE EXPLANATION AND DESCRIPTION OF ALL FIXTURES, DEVICES, DEVICES AND ITEMS OF EQUIPMENT, SHALL BE SUBMITTED FOR APPROVAL.

C. FIELD INSTALLATION DRAWINGS AS REQUIRED TO EXPLAIN FULLY ALL PROCEDURES INVOLVED IN ERECTING, MOUNTING AND CONNECTING ALL ITEMS OF EQUIPMENT.

D. NO EQUIPMENT SHALL BE FABRICATED, DELIVERED, ERECTED, OR RECONNECTED THAN FROM DRAWINGS APPROVED BY THE ENGINEER. SHOP DRAWING IN THE NUMBER DIRECTED SHALL BE SUBMITTED FOR THE FOLLOWING:

1. LIGHTING FIXTURES AND CONTROL DEVICES
2. WIRING DEVICES AND PLATES.
3. CONDUIT, BOXES, AND FITTINGS.
4. WIRE AND CABLE
5. SAFETY SWITCHES AND MOTOR STARTERS
6. FIRE ALARM SYSTEM INCLUDING BATTERY CALCULATIONS AND WIRING DIAGRAMS.

E. IT SHALL BE UNDERSTOOD THAT APPROVAL OF DRAWINGS WILL NOT BIND THE ENGINEER OR THE OWNER TO THE FINAL ACCEPTANCE OF SUCH EQUIPMENT AS THE COMPLETED INSTALLATION AND TEST OF EQUIPMENT AS A WHOLE MUST BE PROVIDED AND GUARANTEED HEREIN AS SPECIFIED.

1.07 TESTS

A. BEFORE AN APPLICATION FOR THE FINAL ACCEPTANCE OF THE WORK WILL BE CONSIDERED, ALL TESTS DEEMED NECESSARY BY THE ARCHITECT TO SHOW PROPER EXECUTION OF THE WORK SHALL HAVE BEEN PERFORMED AND COMPLETED IN THE PRESENCE OF AN ARCHITECT'S REPRESENTATIVE. SCHEDULE OF ALL TESTING PROCEDURES SHALL BE ARRANGED TO SUIT THE CONVENIENCE OF THE ARCHITECT.

B. ANY DEFECTS OR DEFICIENCIES DISCOVERED IN ANY OF THE ELECTRICAL WORK SHALL BE CORRECTED.

1.08 IDENTIFICATION

A. ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL TYPEWRITTEN DIRECTORIES BEHIND TRANSPARENT PLASTIC COVERS IN METAL FRAMES, IN ALL NEW AND EXISTING PANELS INDICATING TYPE AND LOCATION OF LOAD BEING SERVED BY INDIVIDUAL CIRCUIT BREAKERS.

B. ALL PARTS OF EQUIPMENT, SUCH AS PANELS, JUNCTION BOXES, SAFETY SWITCHES, MOTOR STARTER, CIRCUIT BREAKERS, RECEPTACLES, CONDUCTORS, AND SIMILAR ITEMS SHALL BE IDENTIFIED BY NAME, AT SUPPLY END, "LOAD SUPPLIED", AND AT LOAD END "LOAD SUPPLIED FROM".

PART 2 - PRODUCTS

2.01 EQUIPMENT AND MATERIALS

A. ALL EQUIPMENT AND MATERIALS FOR PERMANENT INSTALLATION SHALL BE THE PRODUCTS OF RECOGNIZED MANUFACTURERS AND SHALL BE NEW.

B. NEW EQUIPMENT AND MATERIALS SHALL:
1. WHERE NORMALLY SUBJECTED TO UNDERWRITER'S LABORATORY INC. LISTING OR LABELING SERVICES, BE SO LISTED OR LABELED.
2. BE WITHOUT BLEMISH OR DEFECT.
3. NOT TO BE USED FOR TEMPORARY LIGHT AND POWER PURPOSES WITHOUT ARCHITECT'S AUTHORIZATION.
4. BE IN ACCORDANCE WITH THE LATEST APPLICABLE N.E.M.A. STANDARD.
5. BE APPROVED BY BUILDING MANAGER OR OWNER.

C. FOR ITEMS WHICH ARE TO BE INSTALLED BUT NOT PURCHASED AS PART OF THE ELECTRICAL WORK, THE ELECTRICAL WORK SHALL INCLUDE:
1. THE COORDINATION OF THEIR DELIVERY.
2. THEIR FIELD MAKE-UP AND INTERNAL WIRING AS MAY BE NECESSARY FOR THEIR OPERATION.

D. ELECTRICAL RACEWAY AND SUPPORTING SYSTEMS SHALL BE FURNISHED AND INSTALLED COMPLETE, WITH ALL MATERIALS, FITTINGS, CONNECTIONS AND ACCESSORIES NECESSARY TO PROVIDE IN EACH INSTANCE A COMPLETE OPERATING INSTALLATION, AS DESCRIBED HEREIN, AS INDICATED ON THE DRAWINGS, AND/OR AS APPROVED BY THE BUILDING MANAGER OR OWNER.

E. THE DRAWINGS ARE DIAGRAMMATIC AND GENERALLY INDICATIVE OF THE WORK TO BE INSTALLED, BUT DO NOT SHOW ALL BENDS, FITTINGS, AND BOXES WHICH MAY BE REQUIRED. THE CONTRACTOR SHALL CAREFULLY INVESTIGATE THE JOB CONDITIONS INCLUDING STRUCTURAL AND FINISH CONDITIONS AFFECTING ALL HIS WORK AND ARRANGE THE SAME ACCORDINGLY, FURNISHING SUCH FITTINGS, BOXES AND SIMILAR ITEMS AS MAY BE REQUIRED TO MEET SUCH CONDITIONS.

2.02 WIRING MATERIALS

A. WIRE AND CABLE SHALL BE COPPER, RATED FOR 600 VOLTS, TYPE THHN FOR BRANCH CIRCUITS AND XHHW FOR FEEDERS.

B. WIRE #10 AWG AND SMALLER SHALL BE SOLID, WIRE #8 AWG AND LARGER SHALL BE STRANDED.

C. WIRING SHALL BE CONSISTENTLY COLOR CODED THROUGHOUT. FOR 120/208 VOLT SYSTEMS, UTILIZE RED, BLUE, BLACK FOR LINE (PHASE) CONDUCTORS AND WHITE FOR NEUTRAL CONDUCTOR, SWITCH LEG SHALL BE SEPARATELY IDENTIFIED. GROUND CONDUCTOR SHALL BE GREEN. FOR 277/480V SYSTEMS, UTILIZE BROWN, ORANGE, YELLOW FOR LINE (PHASE) CONDUCTORS, AND GREY FOR NEUTRAL CONDUCTOR. GROUND CONDUCTOR SHALL BE GREEN WITH YELLOW TRACER.

D. MINIMUM SIZE:
1. LIGHTING AND POWER: #12 AWG, UNLESS OTHERWISE INDICATED.
2. CONTROL: #14 AWG.
3. 120 VOLT CIRCUITS OVER 100 FEET IN LENGTH AND 277 VOLT CIRCUITS OVER 200 FEET IN LENGTH FROM THE POINT OF SUPPLY TO THE FIRST OUTLET SHALL BE #10 AWG.

E. SPLICES IN BRANCH CIRCUIT WORK SHALL BE MADE BY MEANS OF TYPE 'R' "SCOTCHLOCK" ELECTRICAL TYPE.

F. ELECTRICAL INSULATION TAPE SHALL BE VINYL PLASTIC TYPE WITH PRESSURE ADHESIVE "SCOTCH" ELECTRICAL TYPE.

G. BRANCH CIRCUITRY NUMBERS INDICATED ON THE DRAWINGS ON MULTI-CIRCUIT HOMERUNS ARE FOR IDENTIFICATION OF DEVICES OR EQUIPMENT THEY ARE CONNECTED TO AND DO NOT NECESSARILY REFER TO PANELBOARD CIRCUIT NUMBERS. ASSIGNMENT OF BRANCH CIRCUIT NUMBERS SHALL BE PART OF THIS WORK AND INDICATED ON PANEL DIRECTORIES. BRANCH CIRCUITS SHALL BE CONNECTED TO CIRCUITS ON PANELBOARDS SO AS TO SECURE A REASONABLE BALANCE ON THE THREE PHASES. WHERE MORE THAN ONE CIRCUIT WITH A COMMON NEUTRAL IS INSTALLED IN THE SAME CONDUIT, EACH PHASE WIRE SHALL BE CONNECTED TO A DIFFERENT LEG OF THE SYSTEM.

H. ALL CONDUCTORS SHALL BE COLOR CODED THROUGHOUT AND NUMBERED AND TAGGED AT EACH JUNCTION BOX, PULL BOX, PANEL, AND DEVICE WITH SUITABLE FIREPROOF TAGS OR ADHESIVE IDENTIFICATION BANDS.

2.03 CONDUITS AND RACEWAYS

A. LIQUID TIGHT FLEXIBLE, GALVANIZED STEEL CONDUIT WITH CONTINUOUS COPPER BONDING CONDUCTOR, SHALL BE USED FOR CONNECTIONS TO MOTORS AND AT OTHER LOCATIONS WHERE VIBRATION MOVEMENT IS ENCOUNTERED.

B. UNLESS OTHERWISE INDICATED OR SPECIFIED ALL WIRING SHALL BE INSTALLED CONCEALED IN CEILINGS, WALLS, SLABS, PIPE CHASES AND FURRED SPACES WHENEVER POSSIBLE.

C. CONDUIT AND FITTINGS SHALL CONFORM TO LATEST ACCEPTABLE STATE CODE AND ALL OTHER CODES HAVING JURISDICTION.

D. CONDUIT SHALL BE 3/4" TRADE SIZE MINIMUM. UNLESS OTHERWISE INDICATED OR SPECIFIED.

E. ALL CONDUITS WHICH ARE TO REMAIN EMPTY FOR FUTURE INTRODUCTION OF CONDUCTORS SHALL BE PROVIDED WITH A PULL LINE WITH IDENTIFICATION BAND AT BOTH ENDS.

F. STEEL JACKETED METAL CLAD CABLE CAN BE USED FOR LIGHTING AND APPLIANCE BRANCH CIRCUITRY IN VOIDS OF CEILING AND PARTITIONS, PROVIDED THAT THIS TYPE OF WIRING IS ACCEPTABLE TO THE LOCAL BUILDING OFFICIAL OR HIS REPRESENTATIVE.

G. RIGID GALVANIZED STEEL (RGS) CONDUIT SHALL CONFORM TO UL 6. FITTINGS SHALL BE THREADED.

H. INTERMEDIATE METALLIC CONDUIT (IMC) SHALL CONFORM TO UL 1242. FITTINGS SHALL BE THREADED.

I. ELECTRICAL METALLIC TUBING (EMT) SHALL CONFORM TO UL 797. FITTINGS SHALL BE GLAND AND RING COMPRESSION TYPE.

J. FLEXIBLE METALLIC CONDUIT SHALL CONFORM TO UL1. LIQUIDTIGHT FLEXIBLE METAL CONDUIT SHALL CONFORM TO UL 360.

K. ALL CONDUIT FITTINGS AND CONNECTORS SHALL BE STEEL WITH INSULATED THROATS, DIE-FORMED ZINC FITTINGS ARE NOT ACCEPTABLE. BUSHINGS SHALL BE PROVIDED AT ALL CONDUIT TERMINATIONS. BUSHINGS LARGER THAN 1" SHALL BE GROUNDING TYPE. PVC BUSHINGS MAY BE UTILIZED ONLY FOR 3/4" BRANCH CIRCUIT CONDUITS TERMINATING AT PANELBOARDS.

2.04 JUNCTION BOXES

A. JUNCTION BOX AND PULL BOXES SHALL BE PROVIDED WHERE INDICATED OR SPECIFIED AND WHERE NECESSARY TO FACILITATE THE INSTALLATION OF EQUIPMENT OR WIRING.

B. ALL BOXES SHALL BE SIZED IN ACCORDANCE WITH NATIONAL ELECTRIC CODE.

2.05 OUTLET BOXES

A. OUTLET, PULL AND JUNCTION BOXES SHALL BE FABRICATED FROM STEEL AND CONFORM TO UL 50, UL 514, AND NEMA OS1. BOXES FOR INTERIOR LOCATIONS SHALL BE CODE GAUGE, GALVANIZED SHEET STEEL.

B. BOXES SHALL CONTAIN SUITABLE KNOCKOUTS. BARRIERS SHALL BE FURNISHED AS REQUIRED BY CODE.

C. BOXES SHALL BE SIZED AS REQUIRED BY CODE FOR NUMBER AND GAUGE OF CONDUCTORS THEREIN, EXCEPT WHERE NOTED TO BE LARGER, THE MINIMUM BOX SIZE SHALL BE 4" SQUARE BE 1 1/2" DEEP. COVERS GREATER THAN 50LBS. SHALL BE DIVIDED INTO MULTIPLE SECTIONS.

D. WHERE REQUIRED AND APPROVED BY THE ENGINEER, EXTRA DEEP OR EXTRA SHALLOW OUTLET BOXES SHALL BE USED TO FACILITATE THE INSTALLATION OF THE CONDUIT SYSTEM.

2.07 WIRING DEVICES

A. ALL DEVICES SHALL BE SPECIFICATION GRADE, U.L. APPROVED.

B. SINGLE POLE SWITCH, 20 AMP, 120/277 VOLTS. HUBBELL OR APPROVED EQUAL.

C. RECEPTACLE, 20 AMP, 2 POLE, 3 WIRE DUPLEX, 125 VOLT, GROUND TYPE. HUBBELL OR APPROVED EQUAL.

D. COLOR AND TYPE OF ALL DEVICE PLATES SHALL BE APPROVED BY ARCHITECT PRIOR TO PURCHASE/INSTALLATION.

2.07 GROUNDING

A. ALL ENCLOSURES AND NON CURRENT CARRYING METALLIC PARTS OF ELECTRICAL EQUIPMENT, RACEWAY SYSTEMS AND EQUIPMENT GROUND BUSES SHALL BE EFFECTIVELY GROUNDED TO THE BUILDING GROUNDING SYSTEMS THROUGH THE SYSTEM GROUND CONDUCTORS. METALLIC CONDUITS AND OTHER RACEWAYS AND ENCLOSURES FOR CONDUCTORS SHALL BE METALLIC ALLOY JOINED TOGETHER INTO A CONTINUOUS ELECTRICAL CONDUCTOR, AS TO PROVIDE EFFECTIVE ELECTRICAL CONTINUITY.

B. GROUND CONTINUITY SHALL BE MAINTAINED THROUGHOUT.

2.09 LIGHTING FIXTURES

A. ALL LIGHTING FIXTURES SHALL COMPLY WITH THE STATE ELECTRIC CODE AND SHALL BE U.L. APPROVED.

B. ALL LIGHTING FIXTURES SHALL BE APPROVED PRIOR TO PURCHASE.

C. ALL LIGHTING FIXTURES SHALL BE FURNISHED AND INSTALLED COMPLETE WITH NECESSARY COMPONENTS, ACCESSORIES, AND LAMPS OF CORRECT TYPE AND RATING AS INDICATED ON ELECTRICAL DRAWINGS.

D. FIXTURES SHALL BE CAREFULLY SUPPORTED AND ALIGNED WITH NECESSARY HANGERS, SUPPORTING MEMBERS, AND FRAMES FOR PROPER INSTALLATION, ALL AS REQUIRED AND AS APPROVED.

E. ALL FIXTURES SHALL BE PROPERLY WIRED AND CONNECTED TO BRANCH CIRCUITS, TESTED AND LEFT READY FOR OPERATION.

F. FOR TYPE, MAKE, AND QUANTITIES OF LIGHTING FIXTURES REQUIRED, SEE ELECTRICAL DRAWINGS AND SPECIFICATIONS.

G. FIXTURES AND/OR FIXTURE OUTLET BOXES SHALL BE PROVIDED WITH HANGERS TO ADEQUATELY SUPPORT THE COMPLETE WEIGHT OF THE FIXTURE. PROVIDE SUPPLEMENTAL SUPPORTS PER ALL LOCAL AND STATE CODES.

2.10 CIRCUIT BREAKERS

A. CIRCUIT BREAKERS FOR PANEL OR INDIVIDUAL MOUNTING SHALL BE MOLDED CASE TYPE, QUICK-MAKE, QUICK-BREAK, OR MANUAL OR AUTOMATIC OPERATION.

B. AMPERE RATING AND NUMBER OF POLES SHALL BE AS INDICATED ON THE DRAWINGS.

C. BREAKERS SHALL BE TYPE AS MANUFACTURED BY THE EXISTING PANELBOARD MANUFACTURER MINIMUM CIRCUIT BREAKER INTERRUPTING CAPACITY SHALL MATCH EXISTING SYSTEM, BUT IN NO CASE SHALL BE LESS THAN 22,000 SYM RMS AMPERES.

2.11 CARTRIDGE FUSES

A. CARTRIDGE FUSES SHALL BE AS FOLLOWS:

1. PROVIDE A COMPLETE SET OF FUSES FOR EACH ITEM OF FUSIBLE TYPE EQUIPMENT. FUSIBLE EQUIPMENT FURNISHED BY OTHER CONTRACTORS WILL BE COMPLETE WITH FUSES.
2. SECONDARY SYSTEM FUSES, RATED AT 600 VOLTS OR LESS, SHALL BE U.L. LISTED AND CONSTRUCTED IN CONFORMANCE WITH THE APPLICABLE STANDARDS SET FORTH BY NEMA AND ANSI. ALL FUSES OF A PARTICULAR CLASS SHALL BE OF SAME MANUFACTURER.
3. REGARDLESS OF ACTUAL FAULT CURRENT, THEY SHALL, AT FULL RECOVERY VOLTAGE, BE CAPABLE OF SAFELY INTERRUPTING FAULT CURRENTS OF 200,000 AMPERES RMS SYMMETRICAL, OR 340,000 AMPERES RMS ASYMMETRICAL, DELIVERABLE AT THE LINE SIDE OF THE FUSE.
4. CIRCUITS 0-600 AMPERES SHALL BE PROTECTED BY THE EQUAL OF BUSSMAN "LOW PEAK" CURRENT LIMITING FUSES, LPN-RKN (250 VOLTS), LPS-RK (600 VOLTS), U.L. CLASS RK-1.
5. FUSES SHALL BE SUITABLE FOR APPLICATION TO FUSE GAPS WHICH REJECT OTHER TYPES OF FUSING.
6. SUPPLY 10% SPARE FUSES OF EACH SIZE AND TYPE 60 AMPS OR LESS. SUPPLY THREE (3) SPARE FUSES FOR EACH SIZE AND TYPE OVER 60 AMPS.

B. CARTRIDGE FUSES SHALL BE MANUFACTURED BY BUSSMAN, GOULD OR EFCO.

2.12 DISCONNECT SWITCHES

A. DISCONNECT (SAFETY) SWITCHES:

1. DISCONNECT (SAFETY) SWITCHES SHALL CONFORM TO INDUSTRIAL STANDARDS OF NEMA, BE UL LISTED AND SHALL BE HEAVY-DUTY TYPE, QUICK-MAKE, QUICK-BREAK TYPE WITH INTERLOCKING COVER MECHANISM AND PROVISIONS FOR PADLOCKING SWITCH HANDLE IN "OFF" POSITION. THREE (3) POLE TOGGLE SWITCHES ARE NOT ACCEPTABLE AS A SUBSTITUTE FOR DISCONNECT SWITCHES.
2. DISCONNECT SWITCHES SHALL BE OF FUSED OR UN-FUSED TYPE AS INDICATED WITH NUMBER DISCONNECTING POLES INDICATED. THE GROUNDED CONDUCTOR SHALL NOT BE SWITCHED. SWITCHES SHALL BE FOR USE WITH CURRENT LIMITING FUSES WITH REJECTION TYPE FUSE CLIPS AND THOSE SHALL BE HORSEPOWER RATED.
3. ENCLOSURES SHALL BE OF PROPER NEMA TYPE FOR THE INTENDED LOCATION AND SHALL BE PHOSPHATE COATED OR EQUIVALENT CODE GAUGE GALVANIZED SHEET STEEL WITH GRAY BAKED ENAMEL FINISH.
4. ACCEPTABLE MANUFACTURERS:
   a. EATON
   b. SIEMENS
   c. SQUARE D
   d. GENERAL ELECTRIC

B. COMBINATION STARTER/DISCONNECT SWITCH: PROVIDE COMBINATION STARTER/DISCONNECT SWITCHES WHERE INDICATED.

2.13 FIRE ALARM EXISTING EQUIPMENT

A. THE EXISTING FACILITY IS EQUIPPED WITH A FIRE ALARM SYSTEM WHICH IS TO BE EXPANDED IN THE WORK AREA. ALL NEW EQUIPMENT SHALL BE OF THE SAME MANUFACTURER AS OF THE EXISTING SYSTEM, INCLUDING PULL STATIONS, SMOKE DETECTORS, DUCT SMOKE DETECTORS, AND AUDIBLE/VISUAL SIGNALS, ETC. ALL NEW HORN/STROBES SHALL BE SYNCHRONIZED WITH EXISTING APPLIANCES.

B. IT SHALL BE THE RESPONSIBILITY OF THE ELECTRICAL CONTRACTOR TO REVIEW THE SCOPE OF WORK AND PROPOSED EQUIPMENT WITH THE SUPERINTENDENT OF FIRE ALARMS FOR THE CITY PRIOR TO PURCHASE AND INSTALLATION. THE E.C. SHALL ALSO NOTIFY THE LOCAL FIRE DEPARTMENT AND THE OWNER AT LEAST 48 HOURS IN ADVANCE OF ANY MODIFICATIONS, POSSIBLE DISRUPTION TO, OR ASSOCIATED WORK ON THE EXISTING FIRE ALARM SYSTEM.

C. THE EXISTING FIRE ALARM SYSTEM SHALL BE REPROGRAMMED AS REQUIRED TO INDICATE THE RENOVATED AREAS, REVIEW ZONING WITH THE OWNER'S REPRESENTATIVE AND LOCAL FIRE MARSHALL. THE E.C. SHALL BE RESPONSIBLE FOR ALL COSTS ASSOCIATED WITH THIS WORK.

D. THE E.C. SHALL BE RESPONSIBLE FOR ALL POWER SUPPLIES REQUIRED DUE TO THE NEW DEVICES/APPLIANCES PROVIDED UNDER THIS CONTRACT.

E. THE E.C. SHALL FURNISH BATTERY CALCULATIONS WHICH INCORPORATE ALL EXISTING EQUIPMENT IN ADDITION TO ALL NEW APPLIANCES/DEVICES.

F. PROVIDE ALL FIRE ALARM WIRING IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

G. ALL FIRE ALARM WIRING SHALL BE INSTALLED IN CONDUIT. ALL CONDUIT SHALL HAVE A RED STRIPE PAINTED EVERY 10'-0". ALL JUNCTION BOXES AND COVERS SHALL BE PAINTED RED. FIRE ALARM RATED "MC" CABLE MAY BE USED WHERE RUN CONCEALED IF ACCEPTABLE TO THE LOCAL FIRE DEPARTMENT.

2.14 FIRESTOPPING, SMOKEPROOFING AND WATERPROOFING

A. PROVIDE FIRESTOP OR SMOKESTOP BETWEEN SLEEVES AND CONDUIT MANUFACTURED BY BIO FIRE SHIELD, INC., OR DOW CORNING CORP. AS FOLLOWS:
1. DOW CORNING SILICONE RTV FOAM.
2. DOW CORNING 96-081 RTV SILICONE ADHESIVE SEALANT.
3. MINERAL FIBER BOARD, MATTING AND PUTTY.

B. PROVIDE WATERPROOFING OF ALL MATERIALS WHICH PENETRATE A FLOOR, EXTERIOR WALL SLAB OR ROOF. ALL SLEEVES SHALL EXTEND A MINIMUM OF 3 INCHES ABOVE FLOOR OR ROOF.

PART 3 - EXECUTION

3.01 DEMOLITION AND REMOVAL WORK

A. REMOVE ALL ELECTRICAL EQUIPMENT, WIRING AND OTHER ELECTRICAL WORK AS REQUIRED. DISCONNECT LOAD AND LINE END OF CONDUCTORS FEEDING DEVICES WHICH ARE TO BE REMOVED OR ABANDONED, REMOVE CONDUCTORS NO LONGER IN USE. CUT BACK TO FLOOR, WALL, OR CEILING AND PLUG BOTH ENDS OF CONCEALED CONDUITS MADE OBSOLETE BY THIS ALTERATION. REMOVE EXPOSED OR ABANDONED CIRCUITS AND OUTLETS. REMOVE MATERIAL AND EQUIPMENT AND DISPOSE OF AS DIRECTED.

B. WHEREVER IT IS REQUIRED TO DISCONNECT OR REMOVE ANY PART OF AN EXISTING CIRCUIT, IMMEDIATELY RECONNECT THAT CIRCUIT OR REESTABLISH SERVICE IN THE REMAINING PORTION OF THE CIRCUIT.

C. THE WORK SHALL ALSO INCLUDE THE REMOVAL OF MATERIALS AS DIRECTED. PRIOR TO REMOVING EQUIPMENT AND MATERIAL FROM PROJECT SITE, THE BUILDING MANAGER OR OWNER WILL INSPECT AND ADVISE WHICH ITEMS WILL BE STORED.

D. WHERE EXISTING RECEPTACLES AND/OR SWITCHES ARE LOCATED IN COLUMNS AND/OR EXTERIOR WALLS, AND ARE NOT TO BE REUSED, REMOVE RECEPTACLE AND CAP OUTLET BOX. RECEPTACLES SHOWN ON PARTITIONS TO BE REMOVED SHALL HAVE ALL WIRING AND CONDUIT REMOVED AS WELL.

E. WHERE PRESENT WORK IS DAMAGED IN THE EXECUTION OF THIS CONTRACT, OR WHERE OPEN

MECHANICAL SPECIFICATIONS

PART 1: GENERAL

1.1 DESCRIPTION OF WORK
A. WORK INCLUDED: PROVIDE LABOR, MATERIALS AND EQUIPMENT NECESSARY TO COMPLETE THE WORK OF THIS SECTION.

1.2 REGULATORY REQUIREMENTS
A. COMPLY WITH ALL APPLICABLE FEDERAL AND STATE LAWS, AND ALL LOCAL CODES, BY-LAWS AND ORDINANCES.
B. REQUEST INSPECTIONS FROM AUTHORITIES HAVING JURISDICTION. OBTAIN ALL PERMITS AND PAY FOR ALL FEES AND INSPECTION CERTIFICATES AS APPLICABLE AND/OR REQUIRED. ALL PERMITS AND CERTIFICATES SHALL BE TURNED OVER TO THE OWNERS UPON COMPLETION OF THE WORK.
C. ALL WORK, EQUIPMENT AND MATERIALS SHALL CONFORM TO THE BUILDING CONSTRUCTION RULES AND REGULATIONS.

1.3 SUBMITTALS
A. PROVIDE SHOP DRAWINGS FOR ALL EQUIPMENT, INCLUDING BUT NOT LIMITED TO:
1. PACKAGED ROOFTOP UNITS.

1.4 COORDINATION
A. WORK SHALL BE PERFORMED IN COOPERATION WITH OTHER TRADES ON THE PROJECT AND SO SCHEDULED AS TO ALLOW SPEEDY AND EFFICIENT COMPLETION OF THE WORK.
B. FURNISH TO OTHER TRADES ADVANCE INFORMATION ON LOCATIONS AND SIZES OF ALL FRAMES, BOXES, SLEEVES AND OPENINGS NEEDED FOR THEIR WORK, AND ALSO FURNISH INFORMATION AND SHOP DRAWINGS NECESSARY TO PERMIT TRADES AFFECTED BY THE WORK TO INSTALL SAME.
C. IF ANY HVAC WORK HAS BEEN INSTALLED BEFORE COORDINATION WITH OTHER TRADES SO AS TO CAUSE INTERFERENCE WITH THE WORK OF SUCH TRADES, ALL NECESSARY ADJUSTMENTS AND CORRECTIONS SHALL BE MADE BY THE HVAC TRADES INVOLVED WITHOUT EXTRA COST TO THE OWNERS.
D. PROTECT ALL MATERIALS AND WORK OF OTHER TRADES FROM DAMAGE WHICH MAY BE CAUSED BY THE HVAC WORK AND REPAIR ALL DAMAGES WITHOUT EXTRA COST TO OWNERS.

1.5 MECHANICAL AND ELECTRICAL COORDINATION
A. HEATING, VENTILATION, AND AIR CONDITIONING SUBCONTRACTOR SHALL FURNISH AND INSTALL VARIOUS ELECTRICAL ITEMS RELATING TO THE HVAC EQUIPMENT AND CONTROL APPARATUS. THE ELECTRICAL SUBCONTRACTOR SHALL BE REQUIRED TO CONNECT POWER WIRING TO THIS EQUIPMENT UNLESS NOTED OTHERWISE.
B. THE HVAC AND ELECTRICAL SUBCONTRACTOR SHALL COORDINATE THEIR RESPECTIVE PORTIONS OF THE WORK, AS WELL AS THE ELECTRICAL CHARACTERISTICS OF THE HVAC EQUIPMENT.
C. ALL POWER WIRING AND LOCAL DISCONNECT SWITCHES WILL BE PROVIDED BY THE ELECTRICAL SUBCONTRACTOR FOR THE LINE VOLTAGE POWER. ALL CONTROL AND INTERLOCKING WIRING SHALL BE THE RESPONSIBILITY OF THE HVAC SUBCONTRACTOR.
D. 120 VOLT POWER WIRING SOURCES EXTENDED AND CONNECTED TO HEATING AND VENTILATING CONTROL PANELS, TRANSFORMERS, AND SWITCHES SHALL BE THE RESPONSIBILITY OF THE ELECTRICAL SUBCONTRACTOR. ALL LOW VOLTAGE THERMOSTAT, ZONE VALVE AND ANY SWITCH WIRING SHALL BE THE RESPONSIBILITY OF THE HVAC SUBCONTRACTOR.
E. ALL ELECTRICAL WORK SHALL CONFORM TO THE REQUIREMENTS OF DIVISION 26.
F. ALL STARTERS SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26 EXCEPT THOSE FURNISHED AS AN INTEGRAL PART OF PACKAGED EQUIPMENT. COORDINATE WITH ELECTRICAL CONTRACTOR.

1.6 INSTALLATION REQUIREMENTS
A. THE ARRANGEMENT OF ALL HVAC WORK SHOWN ON THE DRAWINGS IS DIAGRAMMATICAL ONLY AND INDICATES THE MINIMUM REQUIREMENTS OF THE WORK. CONDITION AT THE BUILDING INCLUDING ACTUAL MEASUREMENTS SHALL DETERMINE THE DETAILS OF THE INSTALLATION.

1.7 RECORD DRAWINGS/PROJECT CLOSEOUT
A. PROVIDE RECORD AS-BUILT DRAWINGS AT COMPLETION OF INSTALLATION.

1.8 GUARANTEE/WARRANTY
A. ALL NEW MATERIALS, ITEMS OR EQUIPMENT AND WORKMANSHIP FURNISHED UNDER THIS SECTION SHALL CARRY STANDARD WARRANTY AGAINST ALL DEFECTS IN MATERIAL AND WORKMANSHIP FOR A PERIOD OF NOT LESS THAN ONE (1) YEAR FROM THE DATE OF FINAL ACCEPTANCE OF WORK. ANY FAULT DUE TO DEFECTIVE OR IMPROPER MATERIAL, EQUIPMENT, WORKMANSHIP OR MANUFACTURING DESIGN WHICH MAY DEVELOP WITHIN THAT PERIOD SHALL BE MADE GOOD, FORTHWITH, BY AND AT THE EXPENSE OF THIS CONTRACTOR, INCLUDING ALL OTHER DAMAGES DONE TO AREAS, MATERIALS AND OTHER SYSTEMS RESULTING FROM THIS FAILURE.
B. THIS CONTRACTOR SHALL GUARANTEE THAT ALL NEW ELEMENTS OF THE SYSTEMS MEET THE SPECIFIED PERFORMANCE REQUIREMENTS AS SET FORTH HEREIN OR AS INDICATED ON THE DRAWINGS.

PART 2: PRODUCTS

2.1 DUCTWORK
A. GENERAL: MATERIAL, CONSTRUCTION AND INSTALLATION SHALL MEET THE REQUIREMENTS OF THE MOST RECENT EDITIONS OF THE FOLLOWING STANDARDS AND REFERENCES, EXCEPT FOR MORE STINGENT REQUIREMENTS SHOWN ON THE DRAWINGS:
1. SMACNA HVAC DUCT CONTRUCTION STANDARDS.
2. SMACNA HVAC AIR DUCT LEAKAGE TEST MANUAL
3. NFPA 90A
4. SMACNA GUIDELINES FOR WELDING SHEETMETAL
PROVIDE SUPPORTING AND HANGING DEVICES NECESSARY TO INSTALL THE ENTIRE HVAC SYSTEM INDICATED ON THE DRAWINGS. DUCTWORK SHALL BE FREE FROM VIBRATION UNDER ALL CONDITIONS OF OPERATION. DIMENSIONS SHOWN ON THE DRAWINGS ARE NET INSIDE DIMENSIONS. NO PIPES, CONDUITS, HANGERS, OR ARCHITECTURAL ELEMENTS SHALL PASS THROUGH DUCTWORK.
B. PRESSURE RATINGS:
1. DUCT CONSTRUCTION CLASS 4" AND STATIC PRESSURE RATING 4" POSITIVE OR NEGATIVE SHALL BE SMACNA SEAL CLASS A AND LEAKAGE CLASS 3 FOR ALL VELOCITIES OF 2000 FEET/MIN. OR LESS. ALL DUCTWORK FROM AIR HANDLER MAINS TO INLETS OF VAV TERMINAL BOXES.
2. DUCT CONSTRUCTION CLASS 2" OR LESS AND STATIC PRESSURE RATINGS OF 2" OR LESS POSITIVE AND NEGATIVE SHALL BE SMACNA SEAL CLASS B AND LEAKAGE CLASS 6 FOR ALL VELOCITIES OF 2000 FEET/MIN. OR LESS. ALL DUCTWORK BETWEEN AIR TERMINALS AND DIFFUSERS.
C. SEALING REQUIREMENTS:
1. CLASS A & BLEAKAGE CLASS 6 & 12: GALVANIZED, NON-WELDED ALUMINUM OR NON-WELDED STAINLESS STEEL DUCTWORK TRAVERSE JOINTS SHALL BE MADE WITH SEALING TAPE EQUAL TO HARDCAST MODEL 1902-FR, CORNERS SHALL BE SEALED AS DESCRIBED BY SMACNA. SEAL ALL NON-FLANGED TRAVERSE JOINTS WITH HARDCAST VERSA MODEL 102 OR APPROVED EQUAL. LONGITUDINAL SEAMS SHALL BE SEALED WITH HARDCAST COLD SEAL MODEL 1001.
D. MATERIALS
1. SHEET METAL DUCTS SHALL BE CONSTRUCTED OF G90 GALVANIZED SHEET METAL UNLESS OTHERWISE SPECIFIED.
E. SUPPORT: SPACE HANGERS AS REQUIRED BY SMACNA (8 FT. MAX.) FOR HORIZONTAL ON 8 FT. CENTERS, UNLESS CONCENTRATED LOADINGS REQUIRE CLOSER SPACING. SUPPORT VERTICAL DUCT AT ROOF AND SLAB PENETRATIONS. SUPPORTS FOR DUCTWORK AND EQUIPMENT SHALL BE GALVANIZED UNLESS SPECIFIED OTHERWISE.
F. CONSTRUCTION:
1. NO SHARP METAL EDGES SHALL EXTEND INTO AIR STREAMS. INSTALL DRIVE SLIPS ON AIR-LEAVING SIDE OF DUCT WITH SHEETMETAL SCREWS ON 6" CENTERS. SPIN IN COLLARS SHALL NOT BE USED FOR BRANCH CONNECTIONS IN 3" OR HIGHER PRESSURE CLASS DUCTWORK.
2. 4" PRESSURE CLASS DUCTWORK (ROUND-SINGLE WALL) JOINT LONGITUDINAL SEAMS SHALL BE LOCAL SPIRAL, LOCK LONGITUDINAL OR BUTT WELDED LONGITUDINAL. TRAVERSE JOINTS SHALL BE SLIP TYPE, DRAW BAND JOINTS SHALL BE USED ON LONGITUDINAL SEAM DUCTS ONLY. SEAMS AND JOINTS IN FITTINGS SHALL BE CONTINUOUSLY WELDED. IF COATING IS DAMAGED DURING WELDING, REPAIR JOINTS TO PREVENT CORROSION.
G. FLEXIBLE DUCTWORK:
1. NOT USED.
H. ACOUSTICAL DUCT LINING:
1. PROVIDE 1.5" THICK ACOUSTICAL LINING BY CERTAIN-TEED, KNAUF, OWENS CORNING OR MANVILLE FOR THE FOLLOWING DUCTWORK:
A) DUCTWORK INDICATED AS LINED ON THE DRAWINGS.
2. INCREASE DUCT DIMENSIONS TO ACCOMMODATE LINING WHILE MAINTAINING INSIDE CLEAR DIMENSIONS SHOWN ON THE DRAWINGS.
3. LINING SHALL BE AS FOLLOWS:
A) LOW-PRESSURE DUCTWORK (BELOW 4" S.P.): SHALL BE BLACK, MAT-FACED, 2 LB. DENSITY, FLEXIBLE GLASS TYPE WITH MIN. NRC VALUE OF 0.75 AND MAX. K-FACTOR AT 75°F MEAN OF 0.24.
B) MEDIUM AND HIGH PRESSURE DUCTWORK (ABOVE 4" S.P.): SHALL BE BLACK, MAT-FACED, 3 LB. DENSITY, RIGID BOARD TYPE WITH MIN. NRC VALUE OF 0.75 AND MAXIMUM K-FACTOR AT 75°F MEAN OF 0.23.
4. FOR ALL LINED DUCTWORK AT THE INLETS AND OUTLETS OF FANS OVER 1 HP, AND FOR ALL LINED MEDIUM AND HIGH PRESSURE DUCTWORK (INCLUDING PLENUMS) WITHIN 20 FEET FROM RTU'S AND AHU'S PROVIDE A PERFORATED ALUMINUM (24 GAUGE) OR GALVANIZED LINER (MIN. 28 GAUGE), WITH 28% MIN. FREE AREA ON THE SIDE OF THE LINER EXPOSED TO THE MOVING AIRSTREAM. METAL LINE SHALL BE HELD IN PLACE BY WELDED PINS SPACED NO MORE THAN 12" O.C.
5. MATERIAL AND INSTALLATION SHALL MEET ALL APPLICABLE CODES AND STANDARDS.
6. DUCT LINER SHALL BE INSTALLED WITHOUT INTERRUPTIONS OR GAPS, USING 100% COVERAGE OF ADHESIVE AND MECHANICAL FASTENERS. MECHANICAL FASTENERS SHALL BE WELDED OR SECURED MECHANICALLY TO DUCT ON 12" MAX. CENTERS.
7. CUT LINER TO ENSURE OVERLAPPED AND COMPRESSED LONGITUDINAL JOINTS AT CORNERS. TRANSVERSE JOINTS IN LINER SHALL ABUT PRECISELY. SEAL JOINTS AGAINST FIBER ENTRAINMENT WITH APPROVED ADHESIVE, AS RECOMMENDED BY MANUFACTURER. USE SHEETMENTAL NOSING AT BEGINNING OF LINING (IN DIRECTION OF FLOW) TO PREVENT EROSION.
8. FRICTION COEFFICIENT CORRECTION FACTOR AT 1000 FPM SHALL BE NO GREATER THAN 1.1.
9. ANY CUT LINER DUE TO DUCT TAKE-OFFS AND BRANCHES SHALL BE TOTALLY SEALED AT EDGES (WITH SHEETMETAL NOSE PIECES) TO PREVENT ENTRAINMENT OF LOOSE FIBERS.
10. DO NOT INSULATE LINED DUCTS.

PART 3: EXECUTION

3.1 TESTING AND CLOSEOUT
A. CLEANING
1. UPON COMPLETION ALL SHEET METAL WORK SPECIFIED UNDER THIS SECTION IS TO BE CLEANED. ALL EQUIPMENT IS TO BE CLEANED, ALL TRIM INSTALLED, ALL PROTECTIVE OIL, TAPE OR OTHER MATERIALS USED TO PROTECT WORK ARE TO BE REMOVED.
2. ALL DUCTS, FANS, AND EQUIPMENT SHALL BE THOROUGHLY CLEANED INSIDE AND OUTSIDE AND BLOWN OUT TO PREVENT AND DEBRIS FROM DAMAGING FAN SHIELDS OR DEBRIS HANGING THROUGH REGISTERS OR DIFFUSERS WHEN SYSTEMS ARE PLACED IN OPERATION. ALL TEMPORARY CONNECTIONS REQUIRED FOR BLOWING OUT THE SYSTEMS, CHEESECLOTH FOR ALL DUCT OPENINGS, AND ANY OTHER EQUIPMENT OR LABOR FOR CLEANING, SHALL BE PROVIDED BY THE HVAC CONTRACTOR. THE ENTIRE HVAC SYSTEM SHALL BE KEPT CLEAN UNTIL FINAL ACCEPTANCE. ANY DAMAGE TO CEILINGS BY THE HVAC CONTRACTOR SHALL BE RECTIFIED BY HIM AT NO ADDITIONAL CHARGE TO THE OWNER, TO THE SATISFACTION OF THE DESIGNER.

3.2 OPERATING AND MAINTENANCE INSTRUCTIONS
A. PRIOR TO COMPLETION OF THE CONTRACT, PROVIDE FIELD AND WRITTEN OPERATING INSTRUCTIONS TO THE OWNER'S DESIGNATED REPRESENTATIVE WITH RESPECT TO OPERATION FUNCTIONS AND MAINTENANCE PROCEDURES FOR ALL EQUIPMENT AND SYSTEMS INSTALLED.
B. PRIOR TO SCHEDULING THE PROJECT FINAL INSPECTION AND AFTER COMPLETION OF ALL INSTALLATION AND RUNNING EQUIPMENT AND AUTOMATIC CONTROL ADJUSTMENTS, PERFORM AIR BALANCING AND ANY OTHER WORK REQUIRED TO PLACE THE EQUIPMENT IN COMPLETE OPERATING CONDITION TO MEET ALL REQUIREMENTS UNDER THIS SPECIFICATION. DURING THIS RUNNING TEST PERIOD, DELIVER TO THE DESIGNER TWO COMPLETE SETS OF OPERATING, SERVICE, MAINTENANCE AND REPLACEMENT DATA FOR ALL EQUIPMENT WHICH WILL REQUIRE OPERATING MAINTENANCE OR REPLACEMENT AND ONE COPY OF THIS LITERATURE SHALL BE AVAILABLE DURING THE INSTRUCTION OF THE OPERATING PERSONNEL WHILE THE OTHER IS CHECKED FOR COMPLETENESS BY THE DESIGNER. DURING ALL WORKING HOURS OF THE "OPERATING TEST", THIS CONTRACTOR'S PERSONNEL SHALL BE AVAILABLE FOR GIVING FIELD INSTRUCTION, SHALL COVER OPERATION, MAINTENANCE AND ADJUSTING OF ALL EQUIPMENT INSTALLED.

3.3 HOISTING, SCAFFOLDING, STAGING AND PLANKING
A. PROVIDE, SETUP AND MAINTAIN ALL REQUIRED DERRICKS, HOISTING MACHINERY, SCAFFOLDS AND STAGING, PLANKING AND PERFORM ALL HOISTING REQUIRED TO COMPLETE THE WORK OF THIS FILED SUB-BID AS INDICATED AND SPECIFIED.
B. SCAFFOLDS SHALL HAVE SOLID BACKS AND FLOORS TO PREVENT DROPPING MATERIALS FROM THERE TO THE FLOORS OR GROUND.

3.4 TESTING AND BALANCING
A. TOTAL SYSTEM BALANCE SHALL BE PERFORMED IN ACCORDANCE WITH AABC NATIONAL STANDARDS FOR FIELD MEASUREMENT AND INSTRUMENTATION, TOTAL SYSTEM BALANCE OR ASHRAE SYSTEMS HANDBOOK.

NEW PACKAGED ROOFTOP GAS FIRED AIR CONDITIONING UNIT SCHEDULE

GENERAL			PERFORMANCE													ELECTRICAL				PHYSICAL		REMARKS			
				COOLING					EVAPORATOR FAN			HEATING (NATURAL GAS)													
TAG	SERVES	LOCATION	NOMINAL TONS	TOTAL MBH	SENSIBLE MBH	EER	IEER	STAGES	CFM	ESP (IN WG)	FLA	INPUT (MBH)	OUTPUT (MBH)	STAGES	THERMAL EFF. (%)	OUTSIDE AIR (CFM)	MCA	MOCP	VOLTAGE	PHASE	WEIGHT (LBS)	MANUFACTURER MODEL	TYPE	RATINGS	FEATURES	INSTALL
RTU-3	TENANT SPACE	ROOF	12.5	150.76	112.09	10.8	15.0	2	5000	1.0	5.6	250.0	205.0	2	82	1000	34	40	460	3	1262	CARRIER 48FEFM14A3A6-8F0A0	(1)	(1)(2)(3)	[1][2]	(1)(2)(3)(4)(5)

(1) NEW PACKAGED ROOFTOP HVAC UNIT, STANDARD EFFICIENCY, NATURAL GAS-FIRED, VERTICAL SUPPLY, VERTICAL RETURN, R-454B

(1) RATINGS IN ACCORDANCE TO ARI STANDARD 210
(2) RATINGS IN ACCORDANCE TO ARI STANDARDS 360 & 370
(3) COOLING PERFORMANCE AT: 91°F ODB, 79°F EDB, 66°F EWB, NOMINAL CFM

[1] HIGH STATIC - ECOBLUE VANE AXIAL FAN, ELECTRO-MECHANICAL CONTROLS WITH POL224, ENTHALPY ECONOMIZER WITH BAROMETRIC RELIEF, R-454B DISSIPATION BOARD AND LEAK SENSOR
[2] 7-DAY PROGRAMMABLE T-STAT WITH SAME NUMBER OF HEATING AND COOLING STAGES AS THE UNIT IT SERVES

(2) DISCONNECT SWITCHES & CONVENIENCE OUTLETS SHALL BE FURNISHED & INSTALLED BY E.C.
(3) PROVIDE CONDENSATE DRAIN TRAP
(4) PROVIDE WITH UNIT DESIGNATION DECAL WITH MINIMUM 4" HIGH REFLECTIVE LETTERS
(4) UTILIZE ORIGINAL ROOF CURB.
(5) UNIT WEIGHT INCLUDES THE EXISTING ROOF CURB WEIGHT.

MECHANICAL LEGEND

SYMBOL	DESCRIPTION
	NEW DUCTWORK, PIPING, EQUIPMENT (LINE WEIGHT)
	EXISTING DUCTWORK, PIPING, EQUIPMENT (LINE TYPE)
	EXISTING HVAC TO BE REMOVED OR RELOCATED
	RECTANGULAR/ROUND SUPPLY AIR DUCTWORK UP
	RECTANGULAR/ROUND RETURN AIR DUCTWORK UP
	RECTANGULAR/ROUND SUPPLY AIR DUCTWORK DOWN
	RECTANGULAR/ROUND RETURN AIR DUCTWORK DOWN
SD	DUCT SMOKE DETECTOR
T	THERMOSTAT
DN	DOWN
E.C.	ELECTRICAL CONTRACTOR
M.C.	MECHANICAL CONTRACTOR
NTS	NOT TO SCALE
RA	RETURN AIR
RTU	ROOFTOP UNIT
SA	SUPPLY AIR

TAG LEGEND

EQUIPMENT (REQUIRING POWER)	EQUIPMENT DESIGNATION / NUMBER

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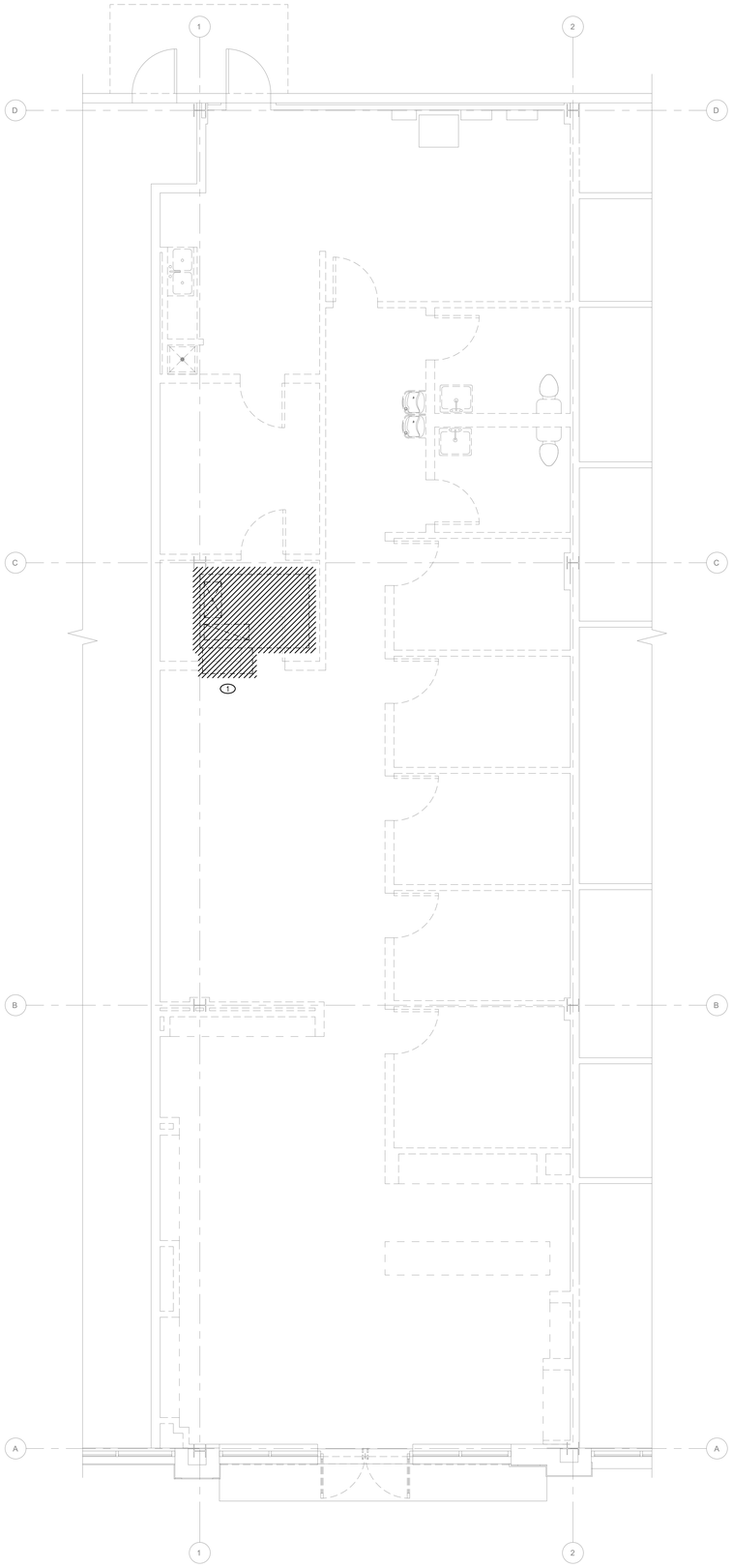
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DAVID R. KNIGHT
MECHANICAL
No. 50220
REGISTERED PROFESSIONAL ENGINEER

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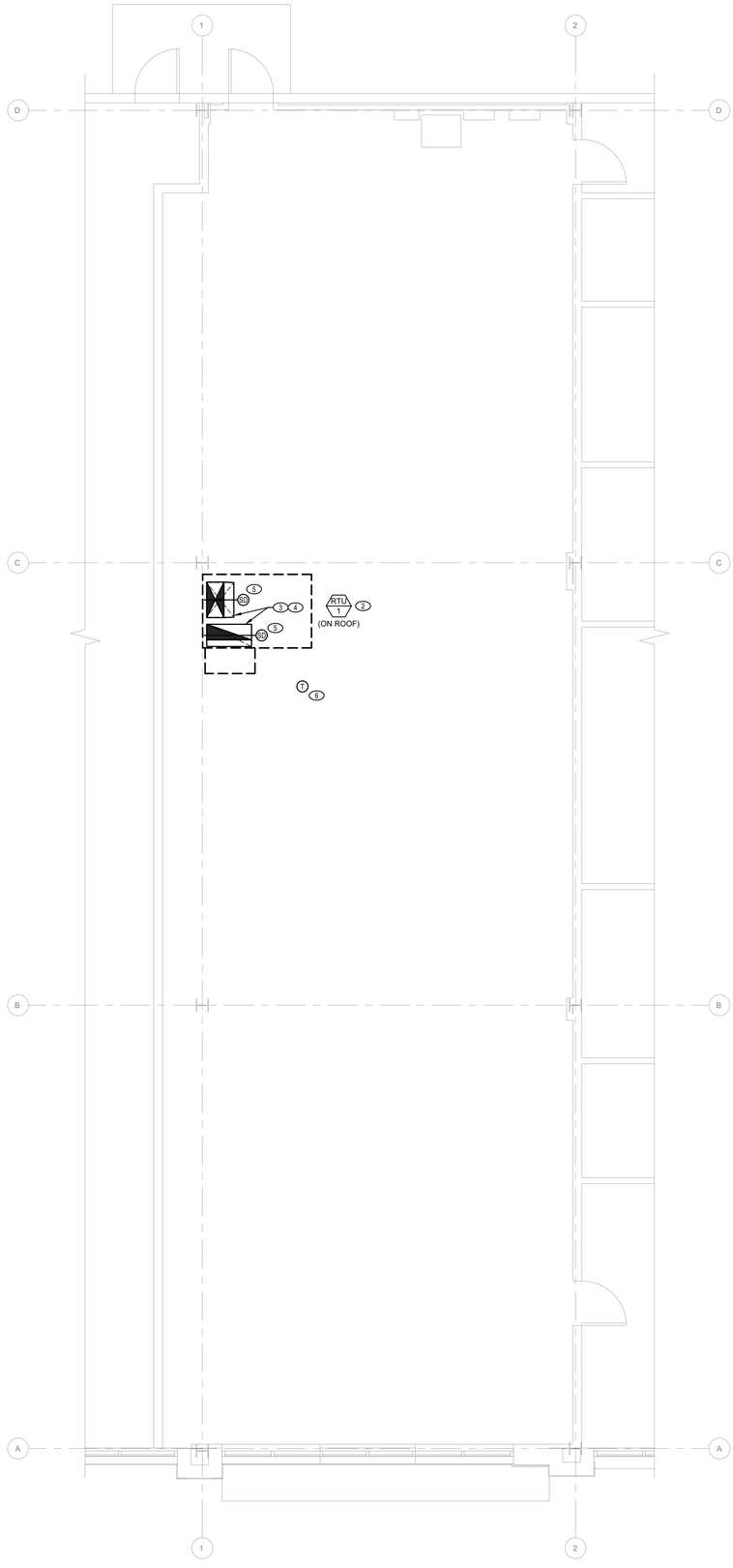
MECHANICAL LEGEND, NOTES, & SCHEDULES

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1 MECHANICAL DEMOLITION PLAN
M - 101
1/4" = 1'-0"



2 MECHANICAL FLOOR PLAN
M - 101
1/4" = 1'-0"

KEY NOTES

1. EXISTING CARRIER RTU ON ROOF (MODEL # 48TCDD08J2M6A6F0JO, SERIAL # 4419P85449) TO BE REMOVED. EXISTING ROOF CURB SHALL REMAIN.
2. PROVIDE NEW ROOFTOP UNIT ON EXISTING ROOF CURB.
3. TRANSITION 28x22 SA & 36x18 RA UP TO RTU-1 ON ROOF, EXTEND DN INTO TENANT SPACE TO THE LENGTH NECESSARY TO INSTALL DUCT SMOKE DETECTORS.
4. PROVIDE DUCTWORK WITH 1.5" THICK ACOUSTICAL DUCT LINING. DIMENSIONS SHOWN INDICATE THE CLEAR INSIDE DIMENSIONS AFTER LINING HAS BEEN APPLIED. INCREASE OVERALL DUCT SIZES TO ACCOMMODATE LINING.
5. NEW DUCT SMOKE DETECTORS SHALL BE FURNISHED & WIRED BY E.C. AND INSTALLED BY M.C.
6. PROVIDE A 7-DAY PROGRAMMABLE T-STAT WITH THE SAME NUMBER OF HEATING AND COOLING STAGES AS THE ROOFTOP UNIT IT SERVES. PROVIDE WITH 150 FEET OF CONTROL WIRING AND LOCATE ON BOTTOM CHORD OF JOIST. TENANT SHALL DETERMINE FINAL LOCATION DURING THEIR FIT-UP.

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DAVID KNIGHT
MECHANICAL
No. 50220
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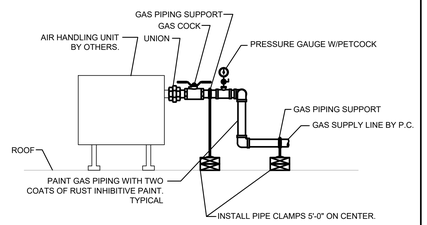
MECHANICAL FLOOR PLANS

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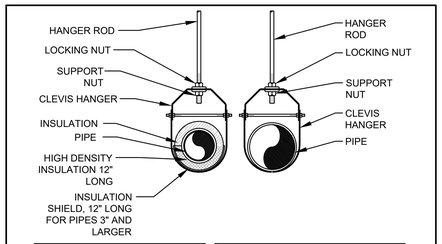
PLUMBING GENERAL NOTES

1. ALL WORK SHALL BE IN STRICT ACCORDANCE WITH ALL LOCAL CODES AND AUTHORITIES HAVING JURISDICTION.
2. THIS CONTRACTOR SHALL VERIFY EXACT LOCATION, SIZE, DEPTH, PRESSURE AND AVAILABILITY OF ALL EXISTING UTILITIES CONCERNED PRIOR TO START OF ANY WORK.
3. THIS CONTRACTOR SHALL COORDINATE WITH ALL OTHER TRADES FOR CLEARANCES AND WORK INCLUDED PRIOR TO START OF WORK.
4. THIS CONTRACTOR SHALL CAREFULLY REVIEW THE EQUIPMENT DRAWINGS FOR NOTES AND DETAILS PERTINENT TO HIS SECTION OF WORK.
5. COORDINATE WITH ELECTRICAL CONTRACTOR PRIOR TO ORDERING EQUIPMENT FOR AVAILABLE VOLTAGES AT EQUIPMENT LOCATIONS.
6. FURNISH NECESSARY PROVISIONS FOR PIPE MOVEMENT WHERE PIPING PENETRATES AN EXPANSION OR SEISMIC JOINT.
7. VERIFY EXACT ROUGH-IN LOCATIONS OF EQUIPMENT WITH EQUIPMENT DRAWINGS.
8. SEE ARCHITECTURAL DRAWINGS FOR EXACT PLUMBING FIXTURE LOCATIONS.
9. VERIFY WITH ARCHITECTURAL DRAWINGS FOR EXACT LOCATIONS OF ALL FLOOR AND AREA DRAINS.
10. PROVIDE CLEANOUTS AND FLOOR CLEANOUTS IN ACCORDANCE WITH CODE. ALL CLEANOUTS SHALL BE ACCESSIBLE.
11. ALL FIXTURES SHALL BE NEW AND PROTECTED FROM ANY DAMAGES DURING CONSTRUCTION.
12. COVER ALL FLOOR DRAINS, FLOOR SINKS, ROOF AND OVERFLOW DRAINS DURING CONSTRUCTION TO PREVENT DEBRIS FROM ENTERING PIPE AND TO PROTECT GRATES FROM DAMAGE.
13. HANDICAPPED USE PLUMBING FIXTURES SHALL BE MOUNTED AT REQUIRED HEIGHT AND CLEARANCES AS PER AMERICANS WITH DISABILITIES ACT.
14. ALL VENTS THROUGH ROOF SHALL BE MINIMUM OF THREE FEET VERTICALLY AND TWENTY FIVE FEET HORIZONTALLY FROM ANY HVAC EQUIPMENT FRESH AIR INTAKES OR BUILDING OPENINGS.
15. INSULATE WASTE PIPE AND 'P' TRAP ABOVE CEILING FROM FLOOR SINKS, FLOOR DRAINS OR FUNNEL DRAINS COLLECTING INDIRECT DRAINS TO PREVENT CONDENSATE DRIPS. INSULATE WASTE PIPE UP TO THE NEXT CONNECTION TO MAIN BRANCH LINE.
16. PROVIDE DRIP PANS BELOW ALL PIPING IN ELECTRIC ROOMS OR OPERATING ROOMS.
17. PROJECT REQUIRES SEISMIC BRACING. REFER TO SPECIFICATIONS.



2 RTU GAS PIPING DIAGRAM

NTS



HANGER ROD SCHEDULE	
PIPE SIZE	ROD SIZE
UP TO 2"	⅜" DIA.
2½" THRU 3"	½" DIA.
4" AND 5"	⅝"
6" THRU 8"	¾"
12"	1"

HANGER ROD SPACING	
PIPE SIZE	MAX. ALLOWABLE SPACING
1"	7'
1 ¼"	8'
1 ½"	9'
2"	10'
2 ½"	11'
3" THRU 8"	12'
10" & 12"	15'

INSULATED AND NON INSULATED PLUMBING SYSTEMS

1 CLEVIS PIPE HANGER/ROD INSTALLATION

NTS

PLUMBING LEGEND

	ETR	LIGHT LINE INDICATES EXISTING PIPING TO REMAIN.
	RE	REMOVE EXISTING PIPING
	CW	COLD WATER
CW	CW	COLD WATER BELOW FLOOR OR BURIED
	HW	HOT WATER
	HWR	HOT WATER RECIRCULATION
	S or W	SOIL OR WASTE
S/W	S or W	SOIL OR WASTE BELOW FLOOR OR BURIED
	V	VENT
V	V	VENT BELOW FLOOR OR BURIED
	CONT	CONTINUATION
	UP	PIPE RISE OR UP
	DN	PIPE DROP OR DOWN
	TEE	PIPE TEE
	SOV	SHUT-OFF VALVE
	PRV	PRESSURE REDUCING VALVE
	CV	CHECK VALVE
	W & T	WASTE & TRAP
	FCO	FLUSH FLOOR CLEANOUT
		ARROW INDICATES DIRECTION OF FLOW
.01		ARROW INDICATES DIRECTION OF SLOPE
		UNION
	BVA	BALANCING VALVE ASSEMBLY
TP	TP	TRAP PRIMER
1 / P-1		DIAGRAM NO. & DWG. NO. REFERENCE
	FD "A"	FLOOR DRAIN & TYPE
WM	WM	WATER METER
	INV	INVERT
	TYP	TYPICAL
	NTS	NOT TO SCALE
	AFF	ABOVE FINISHED FLOOR
	LPC	LIMIT OF PLUMBING CONTRACT
	GC	GENERAL CONTRACTOR
	FPC	FIRE PROTECTION CONTRACTOR
	PC	PLUMBING CONTRACTOR
	EC	ELECTRICAL CONTRACTOR
	S=.01	SLOPE = 1/8" PER FOOT
	S=.02	SLOPE = 1/4" PER FOOT

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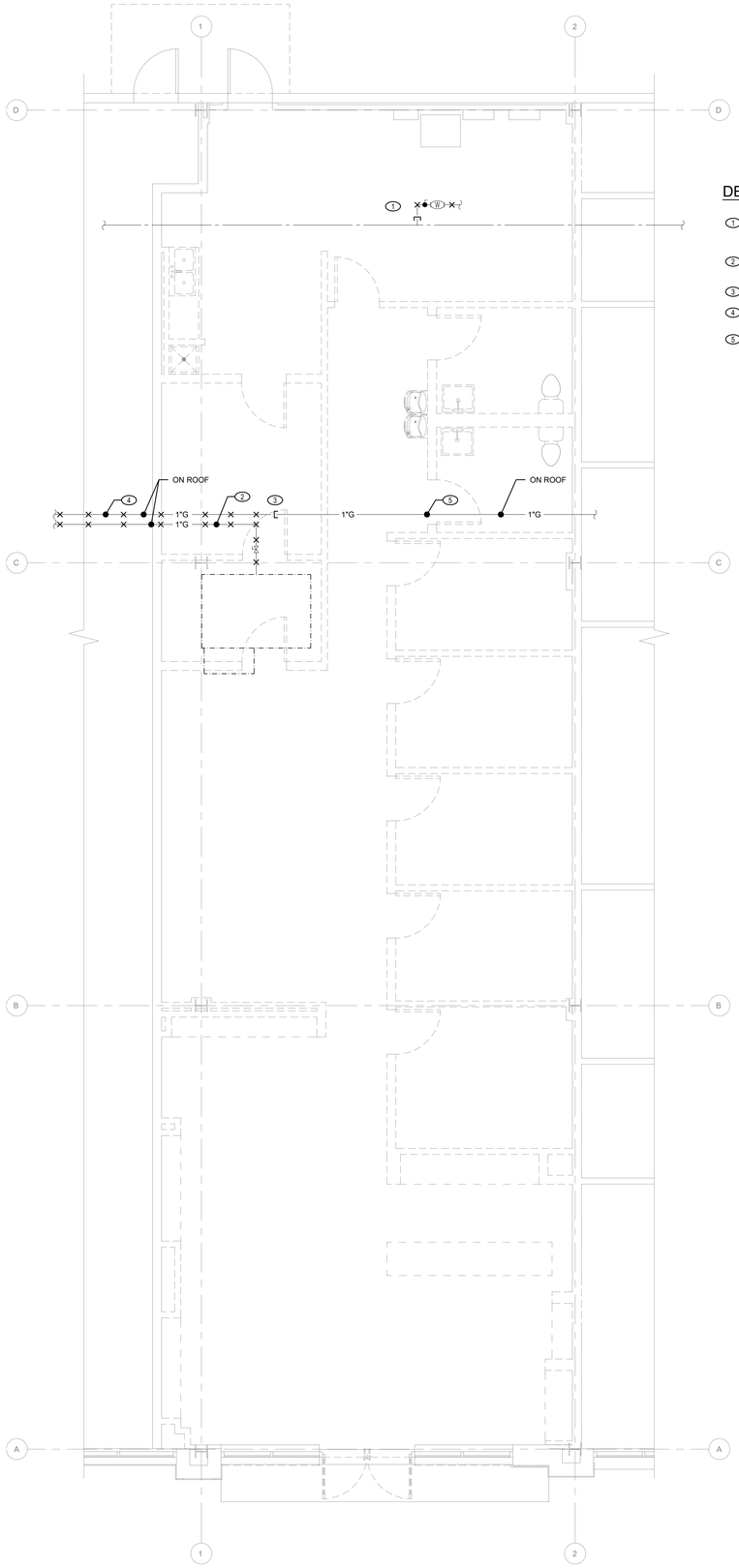
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PLUMBING LEGEND, NOTES, & DETAIL

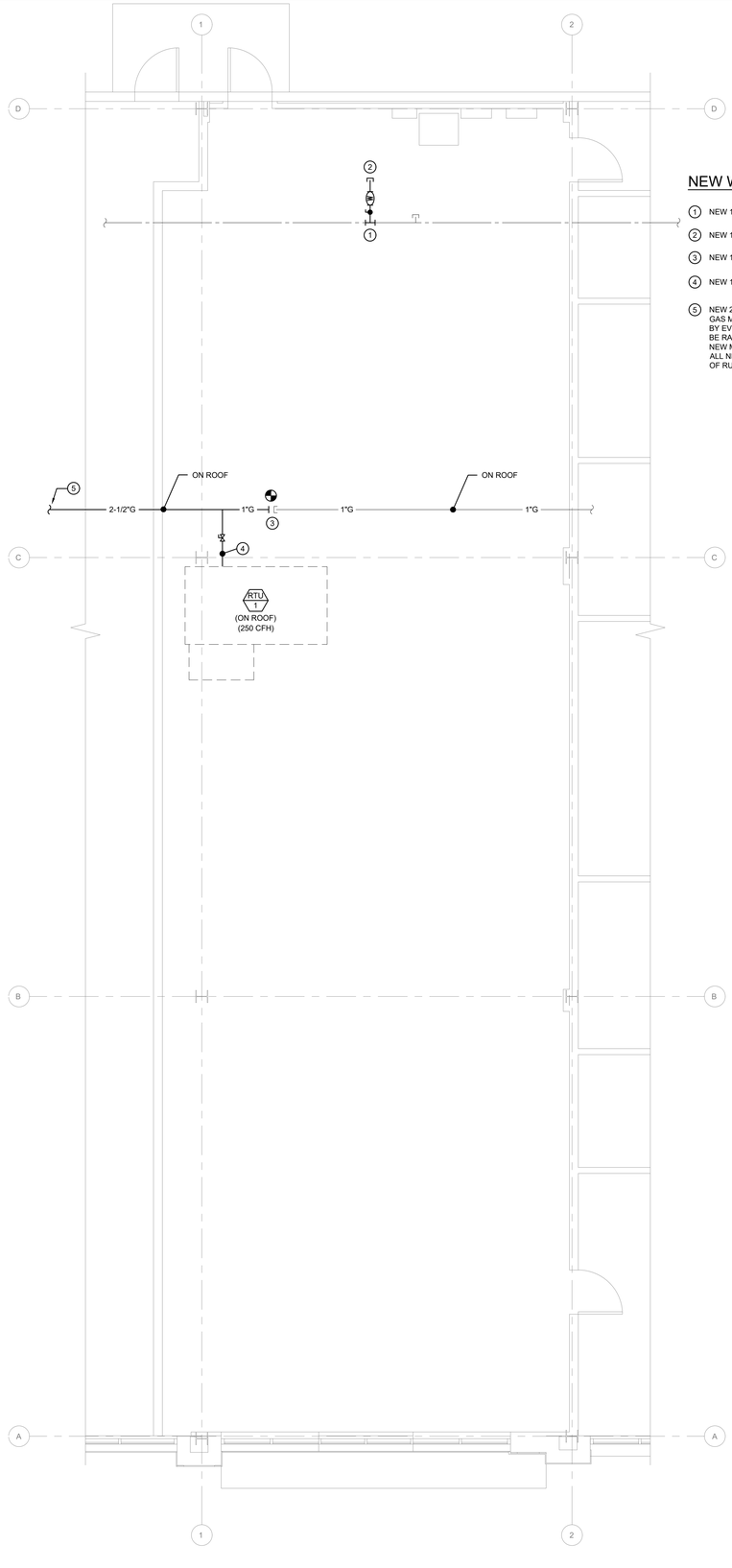
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DEMO KEY NOTES

1. CUT & CAP 1"CW
   REMOVE ALL PIPING DOWNSTREAM
   OF THE WATER METER.
2. DISCONNECT 1"G FROM RTU BEING REMOVED.
   REMOVE 1"G BACK TO GAS METER.
3. CUT & CAP 1"G SERVING ETR RTU'S
4. REMOVE EXISTING 1"G BACK TO EXISTING
   GAS METER.
5. ETR 1"G SERVING EXISTING RTU'S (360 CFH TOTAL)



NEW WORK KEY NOTES

1. NEW 1-1/2"CW, CTE CW MAIN IN CEILING
2. NEW 1-1/2"CW VALVED AND CAPPED W/SUB METER
3. NEW 1-"G, CTE 1"G SERVING EXISTING RTU'S
4. NEW 1-1/4"G TO NEW RTU (250 CFH)
5. NEW 2-1/2"G TO EXTEND TO NEW GAS METER.
   GAS METER TO BE FURNISHED AND INSTALLED
   BY EVERSOURCE. NEW METER CAPACITY TO
   BE RATED FOR 610 CFH. LANDLORD TO COORDINATE
   NEW METER INSTALL WITH EVERSOURCE.
   ALL NEW GAS PIPING TO BE PAINTED W/ 2 COATS
   OF RUST INHIBITIVE PAINT

2 / P - 101 PLUMBING DEMOLITION FLOOR PLAN
1/4" = 1'-0"

1 / P - 101 PLUMBING FLOOR PLAN
1/4" = 1'-0"

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FLOOR PLAN
PLANS

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PLUMBING SPECIFICATIONS

I GENERAL

A. GENERAL: THE WORK COVERED CONSISTS OF FURNISHING ALL LABOR AND MATERIALS NECESSARY TO INSTALL, COMPLETE AND READY FOR CONTINUOUS OPERATION, THE PLUMBING SYSTEMS, APPARATUS AND EQUIPMENT FOR THE CANTON VILLAGE SHOPS RE-DEMISE PROJECT.

B. SHOP DRAWINGS: SHOP DRAWINGS FOR ALL SPECIFIED FIXTURES, EQUIPMENT AND APPARATUS SHALL BE SUBMITTED TO THE ARCHITECT FOR APPROVAL.

C. CODES: ALL EQUIPMENT AND MATERIALS FURNISHED UNDER THE PLUMBING SUB-CONTRACT AND LABOR AND TESTING PERFORMED HEREIN SHALL BE IN COMPLETE ACCORDANCE WITH THE MASSACHUSETTS STATE BUILDING, FUEL GAS, PLUMBING CODES, LOCAL ORDINANCES AND REGULATIONS OF THE CITY OF CANTON, NATIONAL FIRE PROTECTION ASSOCIATION AND INSURANCE REGULATIONS AND REQUIREMENTS GOVERNING SUCH WORK.

D. PERMITS: ANY AND ALL PERMITS REQUIRED FOR INSTALLATION OF ANY MATERIAL SHALL BE OBTAINED AS PART OF THE WORK OF THIS SPECIFICATION INCLUDING ALL FEES OR EXPENSES INCURRED.

E. INSTRUCTIONS: DURING THE ASSEMBLY AND INSTALLATION OF ALL PLUMBING SYSTEMS, THE OWNER'S OPERATING PERSONNEL SHALL BE INSTRUCTED REGARDING ITS OPERATION AND MAINTENANCE. A TWO (2) HOUR INSTRUCTION PERIOD SHALL BE PROVIDED AFTER COMPLETION OF PROJECT. OPERATION AND MAINTENANCE MANUALS SHALL BE REQUIRED.

F. GUARANTEE: ALL MATERIALS AND EQUIPMENT FURNISHED AND INSTALLED UNDER THIS SPECIFICATION SHALL BE GUARANTEED IN WRITING FOR ONE (1) YEAR FROM THE DATE OF ACCEPTANCE OF THE BUILDING BY THE OWNER.

G. RECORD DRAWINGS: THE PLUMBING SUBCONTRACTOR SHALL MAINTAIN AT THE JOB SITE AT ALL TIMES A COMPLETE AND SEPARATE SET OF BLACKLINE PRINTS OF THE PLUMBING DRAWINGS OF HIS TRADE ON WHICH HE SHALL MARK CLEARLY, NEATLY, ACCURATELY, AND PROMPTLY AS THE WORK PROGRESSES. TWO CADD DISKS, AUTOCADD RELEASE 14 OR COMPATIBLE SYSTEM AS WELL AS REPRODUCIBLE AS-BUILTS SHALL BE FURNISHED BY THE PLUMBING SUBCONTRACTOR AT THE JOB COMPLETION.

H. INSPECTION: ALL WORK SHALL BE SUBJECT TO THE INSPECTION OF THE OWNER, THE ARCHITECT AND SUCH OTHER INSPECTORS HAVING JURISDICTION. A PROPERLY EXECUTED CERTIFICATE OF INSPECTION SHALL BE PROVIDED.

I. EXAMINATION OF SITE: THE PLUMBING SUBCONTRACTOR, BEFORE SUBMITTING PRICES OR BEGINNING WORK, SHALL THOROUGHLY EXAMINE THE SITE AND CONTRACT DOCUMENTS. NO CLAIM FOR EXTRA COMPENSATION WILL BE RECOGNIZED IF DIFFICULTIES WHICH AN EXAMINATION OF SITE CONDITIONS AND CONTRACT DOCUMENTS PRIOR TO EXECUTING CONTRACT WOULD HAVE REVEALED.

J. COORDINATION: COORDINATE ALL WORK INSTALLED UNDER THIS SPECIFICATION WITH THAT OF ALL OTHER TRADES.

K. PROTECTION OF PROPERTY: PROTECT ALL NEW AND EXISTING WORK BEFORE, DURING AND AFTER INSTALLATION.

L. DISINFECTION: ALL DOMESTIC WATER SYSTEMS SHALL BE DISINFECTED IN ACCORDANCE WITH THE LOCAL PUBLIC HEALTH AND PLUMBING CODE REQUIREMENTS.

M. TESTS: THE PLUMBING SUBCONTRACTOR SHALL PERFORM ALL TESTS AT THE COMPLETION OF THE WORK, AND THE RESULTS FURNISHED TO THE OWNER AND ARCHITECT IN WRITING.

N. CERTIFICATES OF APPROVAL: UPON COMPLETION OF ALL WORK, THE PLUMBING SUBCONTRACTOR SHALL FURNISH, IN DUPLICATE, CERTIFICATES OF INSPECTIONS FROM ALL INSPECTORS AND AUTHORITIES HAVING JURISDICTION, NOTARIZED LETTERS FROM THE MANUFACTURERS STATING THAT AUTHORIZED FACTORY ENGINEERS HAVE INSPECTED AND TESTED THE INSTALLATION OF THEIR RESPECTIVE SYSTEMS AND FOUND SAME TO BE IN PERFECT OPERATING CONDITION.

O. CONTRACT DRAWINGS: THE CONTRACT DRAWINGS ARE DIAGRAMMATIC AND INDICATE ONLY THE GENERAL ARRANGEMENTS OF WORK. IT IS NOT THE INTENT OF THESE DRAWINGS TO SHOW EVERY PIPE, RISE, DROP, ELBOW, ETC. ANY ADDITIONAL WORK NOT SHOWN AND REQUIRED TO INSTALL THE PLUMBING SYSTEMS SHALL BE INCLUDED AS PART OF THIS CONTRACT.

P. REMOVAL WORK: PARTICULAR CARE SHALL BE TAKEN TO AVOID CREATING HAZARDS ON THE SITE OR CAUSING DISRUPTION OF SERVICE IN THE BUILDING. ALL EXISTING EQUIPMENT TO BE REMOVED SHALL BE DONE IN A NEAT AND WORKMANLIKE MANNER. ALL EXISTING EQUIPMENT TO BE TURNED OVER TO THE OWNER SHALL BE PRESENTED TO THE OWNER IN GOOD CONDITION AT A LOCATION DESIGNATED BY THE OWNER. ALL OTHER EQUIPMENT SHALL BE REMOVED FROM THE PREMISES. REMOVE ALL ABANDONED PIPING AND EQUIPMENT NOT BUILT INTO BUILDING CONSTRUCTION. WHERE CEILING OR WALLS ARE REMOVED, ALL ABANDONED PIPING SHALL BE REMOVED AND ENDS OF LIVE SERVICES CAPPED. ABANDONED ELEMENTS BUILT INTO WALLS OR LOCATED ABOVE EXISTING INACCESSIBLE CEILINGS SHALL REMAIN AND ENDS CAPPED AND MARKED ABANDONED.

Q. CONTINUITY OF SERVICES: SERVICES SHALL BE MAINTAINED IN ALL AREAS WHICH WILL BE OCCUPIED DURING THE CONSTRUCTION PERIOD. WHEN AN INTERRUPTION OF SERVICE BECOMES NECESSARY, SUCH SHALL BE MADE ONLY UPON CONSENT OF THE OWNER AT A TIME OUTSIDE NORMAL WORKING HOURS AS HE SHALL DESIGNATE. REFER TO THE OVERALL SCHEDULING OF THE WORK OF THE PROJECT. SCHEDULE WORK TO CONFORM TO THIS SCHEDULE AND INSTALL WORK TO NOT DELAY NOR INTERFERE WITH THE PROGRESS OF THE PROJECT.

R. SEISMIC RESTRAINTS: INSTALLATION OF PLUMBING EQUIPMENT, ACCESSORIES AND COMPONENTS SHALL BE IN ACCORDANCE WITH THE SEISMIC REQUIREMENTS IDENTIFIED IN THE MASSACHUSETTS STATE BUILDING CODE, NINTH (9TH) EDITION.

II. SCOPE

A. THE WORK OF THIS SECTION CONSISTS OF ALL LABOR, MATERIALS AND EQUIPMENT REQUIRED TO PROVIDE ALL PLUMBING WORK COMPLETE, IN PLACE, AS SHOWN ON THE DRAWINGS, SPECIFIED HEREIN AND AS NECESSARY FOR A PROPER INSTALLATION.
B. THE EXTENT OF THE PLUMBING WORK SHALL INCLUDE, BUT NOT BE LIMITED TO THE FOLLOWING:
1. DOMESTIC COLD WATER SYSTEM.
2. DOMESTIC HOT WATER SUPPLY SYSTEM.
3. SANITARY WASTE AND VENT SYSTEM.
4. ALTERATIONS, ADDITIONS AND/OR REMOVAL OF EXISTING PLUMBING SYSTEMS AND FIXTURES WITHIN THE RENOVATED AREA IN ORDER TO CONFORM TO NEW SPACE REQUIREMENTS.
5. INSTALLATION OF TOILET ROOM ACCESSORIES.
6. CORE DRILLING.
7. FURNISHING OF ACCESS PANELS.

III RELATED WORK

A. THE FOLLOWING EQUIPMENT ITEMS AND WORK SHALL BE THE RESPONSIBILITY OF OTHERS:
1. CUTTING AND PATCHING
2. TEMPORARY WATER, HEAT, FIRE PROTECTION AND TOILET FACILITIES
3. TEMPORARY POWER AND LIGHTING
4. EXCAVATION AND BACKFILL
5. CONCRETE AND MASONRY SUPPORTS AND EQUIPMENT BASES
6. FLASHING AND CAULKING
7. FINISH PAINTING
8. HEATING, VENTILATING AND AIR CONDITIONING
9. FIRE PROTECTION
10. ELECTRICAL POWER AND WIRING
11. FURNISHING OF TOILET ROOM ACCESSORIES
12. INSTALLATION OF ACCESS PANELS
13. FOOD SERVICE EQUIPMENT
14. LABORATORY EQUIPMENT
15. FOUNDATIONS AND TRENCHING
16. ELECTRIC HEAT TRACING

IV MATERIALS

A. PIPE AND FITTINGS

1. TYPE A: (POTABLE AND NON-POTABLE WATER)

TYPE L HARD DRAWN COPPER TUBING WITH WROUGHT COPPER SWEAT FITTINGS JOINED WITH APPROVED 95/5 LEAD FREE TIN ANTIMONY SOLDER.

2. TYPE B: (WASTE AND VENT SMALLER THAN 2")

TYPE DWV HARD DRAWN SEAMLESS COPPER TUBING WITH WROUGHT COPPER DRAINAGE FITTINGS JOINED WITH 95/5 LEAD FREE TIN ANTIMONY SOLDER.

3. TYPE C: (SANITARY, WASTE AND VENT INSIDE BURIED)

SERVICE WEIGHT BELL AND SPIGOT CAST IRON SOIL PIPE AND FITTINGS JOINED WITH NEOPRENE RESILIENT GASKETS.

4. TYPE D: (SANITARY, WASTE AND VENT, STORM)

NO HUB CAST IRON SOIL PIPE AND FITTINGS JOINED WITH APPROVED STAINLESS STEEL MECHANICAL COUPLINGS WITH NEOPRENE GASKETS.

5. PIPE AND FITTINGS SHALL BE IN ACCORDANCE WITH THE FOLLOWING:

a. COLD WATER TYPE A
b. HOT WATER SUPPLY TYPE A
d. SANITARY WASTE AND VENT INSIDE BURIED TYPE C
e. SANITARY WASTE AND VENT TYPE D
f. WASTE AND VENT SMALLER THAN 2" TYPE B

B. INSULATION

1. ALL DOMESTIC COLD AND HOT WATER SUPPLY PIPE, FITTINGS AND VALVES SHALL BE INSULATED WITH HEAVY DENSITY RIGID FIBERGLASS WITH A VAPOR BARRIER AND ALL PURPOSE JACKET WITH SELF-SEALING LAP JOINT. VALVES AND FITTINGS SHALL BE INSULATED WITH ZESTON HI-LO INSULATION AND COVERED WITH 25/50 RATED PVC COVERS SECURED WITH VAPOR RETARDER MASTIC.
2. ALL HORIZONTAL ROOF DRAINAGE PIPE, ROOF DRAIN BODIES AND THE BEND INTO VERTICAL POSITION SHALL BE INSULATED IN THE MANNER DESCRIBED FOR COLD AND HOT WATER.
3. WASTE, COLD WATER AND HOT WATER BENEATH HANDICAPPED LAVATORIES SHALL BE INSULATED WITH BROCAR PRODUCTS SERIES 500 INSULATION KIT FOR P TRAP ASSEMBLY AND HOT WATER AND COLD WATER ANGLE VALVE ASSEMBLY. KIT SHALL BE WHITE FLEXIBLE VINYL INSULATION SECURED WITH NYLON FASTENERS SUPPLIED, FOAM INSERTS, ANTIMICROBIAL.
4. INSULATION THICKNESS SHALL BE AS FOLLOWS:

COLD WATER	=	1"
HOT WATER SUPPLY	=	1"

C. PIPE SLEEVES, HANGERS AND SUPPORTS

1. ALL PIPING SHALL BE PROPERLY SUPPORTED FROM THE BUILDING STRUCTURE IN ACCORDANCE WITH LOCAL CODES AND MANUFACTURER'S RECOMMENDATIONS. HANGERS FOR INSULATED PIPING SHALL BE OVERSIZED AND FURNISHED WITH A SHEETMETAL INSULATION SHIELD TO ALLOW THE INSULATION TO PASS THROUGH UNCUT. PROVIDE SCHEDULE 40 PIPE SLEEVES, EXTEND 1 INCH ABOVE FLOOR, MAKE WATERTIGHT AND PACK WITH MATERIAL THAT SHALL MAINTAIN FIRE RATING. PROVIDE CORE DRILLING WHERE REQUIRED AND PROVIDE FIRE RATED LINK SEAL PENETRATION CLOSURES.

D. VALVES

1. ALL SHUT OFF VALVES ON COLD AND HOT WATER PIPING FROM ½ INCH UP TO AND INCLUDING 2 INCH SHALL BE APOLLO SERIES 77-200, SOLDER END, BRONZE BODY BALL VALVE, CHROME-PLATED BRONZE BALL, 600 PSI WOG, FULL PORT BALL VALVE.
2. ALL SHUT OFF VALVES ON COLD AND HOT WATER PIPING ½ INCH SHALL BE APOLLO SERIES 70-200, SOLDER END, BRONZE BODY BALL VALVE, CHROME-PLATED BRONZE BALL, 600 PSI WOG.
3. ALL CHECK VALVES ON COLD WATER, HOT WATER AND HOT WATER RECIRCULATION PIPING THREE INCHES AND LESS IN SIZE SHALL BE NIBCO FIGURE NO. S-413-W, SOLDER END, BRONZE BODY SWING CHECK, BRONZE DISC, 200 PSI WOG.
4. ALL DRAIN VALVES SHALL BE ½ INCH APOLLO MODEL 78-103 WITH WATTS NO. 8A HOSE CONNECTION VACUUM BREAKER, CAP WITH CHAIN OF LENGTH AS REQUIRED.
5. ALL SHUT-OFF VALVES ON NATURAL GAS SYSTEM 2 INCHES AND SMALLER SHALL BE APOLLO SERIES 70-100-07 MASSACHUSETTS APPROVED, THREADED BRONZE BALL VALVE, 600 PSI WOG.
6. ALL BALL VALVES FOR INSTALLATION IN INSULATED PIPING SHALL HAVE VALVE EXTENSIONS TO SUIT INSULATION THICKNESS.

E. PLUMBING FIXTURES

IN GENERAL, ALL PLUMBING FIXTURES SHALL BE WALL HUNG, WHITE VITREOUS CHINA WITH CHROME-PLATED FAUCETS, STOPS AND TRAPS. ALL SUPPLY STOP VALVES SHALL BE BRASS BODY AND STEM AND HAVE THREADED OR SWEAT SOLDER INLET. PROVIDE WALL ESCUTCHEONS. FIXTURES AND TRIM SHALL BE OF THE SAME MANUFACTURER SIMILAR TO KOHLER, ELJER OR AMERICAN STANDARD.

a. REFER TO THE PLUMBING FIXTURE SCHEDULE ON SHEET P0.1

1. FIXTURES DESIGNATED FOR BARRIER FREE USE SHALL BE MOUNTED IN ACCORDANCE WITH THE AMERICANS WITH DISABILITIES ACT (ADA) AS WELL AS STATE AND LOCAL CODES. WATER CLOSET FLUSH VALVES SHALL BE MOUNTED ON THE WIDE SIDE OF THE TOILET AREAS.

2. ALL LAVATORY CONTROLS, WHERE APPLICABLE, SHALL BE ADJUSTED BY INSTALLING PLUMBER PRIOR TO THE FINAL INSPECTION. CONTROLS SHALL BE SET TO DELIVER WATER AT A MAXIMUM TEMPERATURE OF 110 DEGREES F.

F. PIPE IDENTIFICATION AND VALVE TAGS

1. ALL PLUMBING SYSTEMS SHALL BE LABELED AT EACH VALVE, AT EACH BRANCH, AT EACH PIPE PASSAGE THROUGH WALL AND AT INTERVALS OF NOT MORE THAN 20 FEET WITH COLOR CODED SEMI-RIGID SETMARK PIPE MARKERS WITH ARROWS INDICATING THE DIRECTION OF FLOW. ALL VALVES SHALL BE TAGGED WITH 1-1/2 INCH DIAMETER BRASS TAGS AND NUMBERED IN SEQUENCE FROM POINT OF ORIGIN. VALVE CHARTS SHALL BE PLACED UNDER GLASS, FRAMED AND PRESENTED TO THE OWNER.

G. CLEANOUTS

1. CLEANOUTS SHALL BE IRON BODY WITH HEAVY BRASS PLUG AND RAISED NUT, SAME SIZE AS PIPE FOR PIPING UP TO FOUR INCHES AND NOT LESS THAN FOUR INCHES IN SIZE FOR PIPING LARGER THAN FOUR INCHES AND CLOSED GAS TIGHT. FLOOR CLEANOUTS IN CARPETED AREAS SHALL HAVE CARPET CLEANOUT MARKERS. FLOOR CLEANOUTS SHALL NOT BE LOCATED BENEATH PARTITIONS, CASEWORK, NON-PORTABLE EQUIPMENT OR SIMILAR INSTALLATION CONDITIONS. END CLEANOUTS ON NO HUB CAST IRON SHALL BE JOSAM SERIES 58900-20. END CLEANOUTS ON COPPER WASTE SHALL BE NIBCO 816. FLUSH FLOOR CLEANOUT SHALL BE JOSAM SERIES 56000-2-22-41 IN CONCRETE FLOORS. THE LAST FLUSH FLOOR CLEANOUT BEFORE EXITING THE BUILDING SHALL BE JOSAM SERIES 56010-2-22-41. EXPOSED DANDY CLEANOUTS ON NO HUB CAST IRON SHALL BE JOSAM SERIES 58910-20. WALL CLEANOUTS AND CONCEALED DANDY CLEANOUTS ON NO HUB CAST IRON SHALL BE JOSAM SERIES 58910-19 WITH SERIES 58890 CLEANOUT PLUG WITH CENTER SCREW LENGTH AS REQUIRED. END CLEANOUTS ON POLYPROPYLENE PIPING SHALL BE FUSEAL FITTING CLEANOUT ADAPTER WITH THREADED PLUG.

H. WATER HAMMER ARRESTORS (SHOCK ABSORBERS)

1. MAINTENANCE-FREE WATER HAMMER ARRESTORS SHALL BE FURNISHED AND INSTALLED IN ACCESSIBLE LOCATIONS AT ALL LOCATIONS IN THE WATER SYSTEMS WHERE QUICK ACTING VALVES ARE INSTALLED AS WELL AS WHERE VER HAMMER MAY OCCUR.
2. WACO HAMMER ARRESTORS SHALL BE SIMILAR TO THE FOLLOWING JAY R. SMITH MODEL NUMBERS. SIZING AND PLACEMENT SHALL BE IN ACCORDANCE WITH PDI STANDARD PDI-WH-201 AND THE MANUFACTURER'S RECOMMENDATIONS.

TYPE	FIXTURE UNIT RATINGS	MODEL
SA-A	1-11	JAY R. SMITH 5005
SA-B	12-32	JAY R. SMITH 5010
SA-C	33-60	JAY R. SMITH 5020

I. ACCESS PANELS

FURNISH ACCESS PANELS FOR ACCESS TO ALL CONCEALED PARTS OF THE PLUMBING SYSTEMS THAT REQUIRE ACCESSIBILITY FOR THE PROPER OPERATION AND MAINTENANCE OF THE SYSTEM. SIZE SHALL BE SUFFICIENT FOR THE PURPOSE, BUT NO LESS THAN 12 INCHES BY 18 INCHES. ACCESS DOORS SHALL BE PRIME COATED OF RUST INHIBITIVE PAINT, CONTINUOUS HINGE AND MANUFACTURED BY INLAND STEEL PRODUCTS COMPANY MILCOR.

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